Supplementary materials

Bayesian Brain in ADHD: potential catecholaminergic pathway of volatility estimation

Plank et al.

2026-03-20

Contents

S1 Package versions

The following packages are used in this RMarkdown file:

```
## [1] "R version 4.5.2 (2025-10-31)"

## [1] "knitr version 1.51"
## [1] "ggplot2 version 4.0.2"
## [1] "brms version 2.23.0"
## [1] "designr version 0.1.13"
## [1] "bridgesampling version 1.2.1"
## [1] "tidyverse version 2.0.0"
## [1] "ggpubr version 0.6.2"
## [1] "ggrain version 0.1.0"
## [1] "bayesplot version 1.15.0"
## [1] "SBC version 0.5.0.9000"
## [1] "rstatix version 0.7.3"
## [1] "BayesFactor version 0.9.12.4.7"
```

```
## [1] "effectsize version 1.0.1"
## [1] "bayestestR version 0.17.0"
```

S2 Methods: additional details

S2.1 Sample and power analysis

All participants, ranging in age from 16 to 45 years with an estimated IQ above 70, had sufficient knowledge of German to understand task instructions, normal or corrected-to-normal vision, no neurological diagnoses and no acute suicidality. COMP adults did not take any psychotropic medication, whereas the ADHD and ADHD+ASD groups included participants regardless of such medication use. All individuals in the ADHD and ADHD+ASD groups held formal diagnoses of ADHD or both ADHD and ASD, respectively. Additionally, participants from the ADHD and COMP groups also participated in a separate study involving magnetic resonance imaging (Plank, Yurova, et al., 2025).

Based on a sample size estimation performed in PANGEA v0.2 (Westfall, 2016), our goal was to analyse at least 22 participants per group to achieve a power of 80% to detect a medium between-subjects effect of d = 0.45 in all tasks used in the project. For the PAL task analysed in this study, 66 participants were nested in three groups and 336 trials nested in two predictors, resulting in a power of 94%. We recruited 72 participants for this study. Data of six participants was excluded from analysis (task accuracy below 66.7%: two ADHD and two COMP; suicidality: one ADHD+ASD; change in diagnosis: one ADHD+ASD), resulting in 22 participants per group (see Table 1). However, technical difficulties led to a smaller sample of 57 participants for the pupillometry analysis (17 ADHD, 22 ADHD+ASD, 18 COMP). Of the 44 participants with ADHD in the behavioural sample, nine were taking amphetamines, two were taking atomoxetine and 14 were taking methylphenidate. Of these 25 participants taking medication for ADHD, seven also took other psychotropic medication, with an additional four participants taking no ADHD medication but other psychotropic medication. Thus, only 15 of the participants with ADHD did not take any psychotropic medication.

S2.2 Procedure

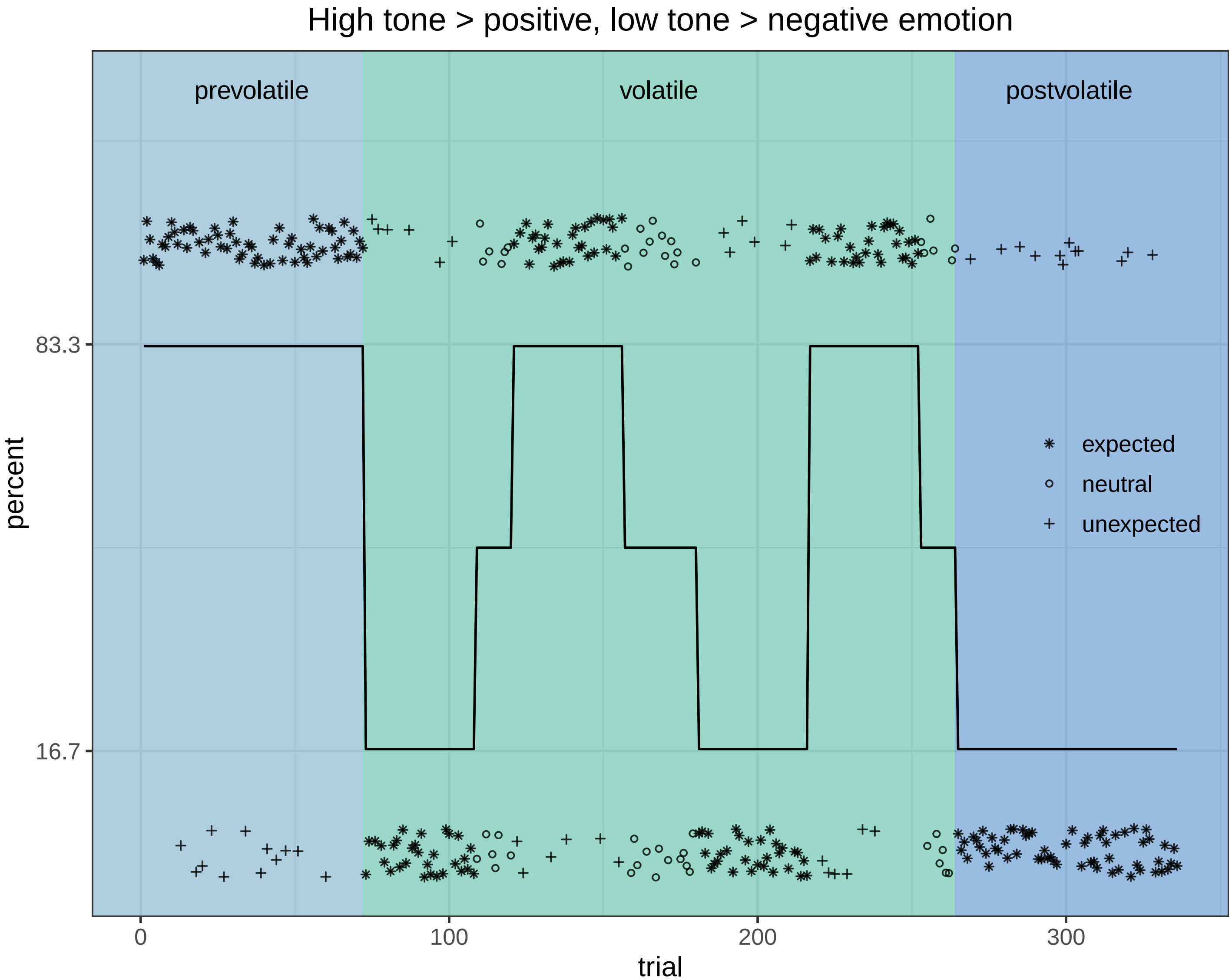
Behavioural tasks were performed in a small, soundproof testing chamber with consistent lighting on a Lenovo Legion Pro 5 mounted in a laptop stand (resolution: 2560x1600 pixel; size: 344x215mm; refresh rate: 60Hz). Eye-tracking data was collected with a LiveTrack Lightning eye tracker (Cambridge RS, sampling rate 500Hz) using a headrest to ensure a stable viewing distance of 57cm. The eye tracker was individually calibrated before each task based on 9-points.

Data collection started with a sociodemographic questionnaire and the BDI (Hautzinger et al., 1994) to assess acute suicidality, which was an exclusion criterion. All participants, who reported no acute suicidality and for whom no IQ estimates were on file from previous testing sessions, performed a verbal and non-verbal IQ assessment (MWT-B, Lehrl, 1999; CFT 20-R, Weiß, 2006). The rest of the testing session consisted of clinical self-assessments (TAS-20, Grabe et al., 2009; ASRS-v1.1, Kessler et al., 2005; RADS-R, Rausch et al., 2024). The TAS-20, measuring alexithymia, was always performed at the end of the testing session to prevent any biasing effects.

S2.3 Stimulus creation

Stimuli were created using KDEF pictures of people portraying three expressions: emotionally neutral, fearful and happy (Lundqvist et al., 1998). Happy and fearful pictures were morphed with emotionally neutral pictures of the same person using WinMorph 3.01 to create videos of the person gradually and continually taking on an emotional expression. Independent samples of 21 and 25 participants rated the naturalness of videos of female-presenting and male-presenting faces, respectively (online surveys on SoSci, Leiner, 2019). We selected six identities based on these ratings (F02, F05, F09, M10, M11 and M13; naturalness of selected videos: range = 70.04-82.81; mean = 76.55). Of these videos, we extracted still images capturing 50%, 75% and 100% of the emotional expression to create hard, medium and easy stimuli, respectively.

S2.4 Development of associations across PAL task



S2.5 Computational modelling of reaction times: Hierarchical Gaussian Filter

To ensure model reliability and faithfulness, we followed the Bayesian workflow adaptation for computational modelling as proposed by Hess and colleagues (2025). This workflow includes the computation of empirical priors, in our case based on the publicly available data from Lawson et al. (2021) as well as our own pilot data, simulation-based evaluation of parameter recovery and model identifiability, posterior predictive checks and model comparison.

S2.6 Comparison of LME across groups

```
##                      bf        error                          time        code
## diagnosis -0.5840522 9.993788e-05 Thu Oct 30 11:19:46 2025 eca1873f721
```

S2.7 Analysis of extracted HGF parameters

We analysed HGF parameters using a Bayesian linear model with a Gaussian distribution and included group as the population-level predictor. Additionally, we explored whether ADHD diagnosis or ADHD medication could be predicted using scaled HGF parameters by fitting two Bernoulli models. For these Bernoulli models, we did not distinguish between adults with ADHD alone and those with comorbid ASD. We used a lognormal Bayesian linear mixed model to assess the influence of alpha level (second, third level), change (prevolatile to volatile, volatile to postvolatile) and their interaction on learning rate updates. The model also includes participants (level and change slopes) on the group-level.

S2.8 Computational modelling of reaction times: Wiener Diffusion Model

We used the ggdmc package to implement a Bayesian version of a hierarchical Wiener diffusion model (Lin & Strickland, 2020), since Bayesian versions of DDM are more robust to designs with fewer trials (Myers et al., 2022). We constructed three separate hierarchical DDMs, one for each group, as a single model for all groups did not converge. Each model included one drift rate per task phase (prevolatile, volatile, postvolatile) to assess differences between task phases with high and low environmental volatility. We assessed model reliability with simulation-based parameter recovery. Only models where all potential scale reduction factors assessing convergence of the Markov chains were smaller than 1.1 were included in the analysis (Gelman et al., 2003). To achieve convergence, we increased the thinning until stable models were obtained. Then, we extracted participant-specific DDM parameters and analysed the drift rates with a Bayesian linear mixed model with a Gaussian distribution (population-level predictors group, phase and their interaction; group-level intercept for participants). A similar model focusing on adults with ADHD, regardless of comorbid ASD, investigated the influence of medication on drift rates.

S2.9 Conventional analysis of reaction time variance, reaction times and error rates

We included only reaction times on correct trials in the analysis of the reaction times and reaction time variances. Additionally, we excluded extreme reaction times defined by the interquartile method. Participants whose accuracy was below 66.67% were excluded from all analyses. Additionally, reaction time variance was only computed as standard deviation for conditions where at least two-thirds of the trials' reaction times were included in the analysis. We used a shifted lognormal distribution to model correct reaction times with the model including group, expectancy, phase and difficulty on the population level. On the group level, we added trials (slope for group) and participants (slopes for expectancy, phase, difficulty and their interactions). We analysed reaction time variance with a lognormal model including the population-level predictors group, expectancy, phase and their interaction as well as participants with slopes for expectancy and phase on the group-level. We dropped the predictor difficulty due to poor posterior predictive fit, slightly deviating from the preregistration. Additionally, we used a Bayesian ANOVA on the rank transformed accuracies to explore differences.

S3 Participant-specific HGF and DDM parameters

S3.1 H3c: second level tonic volatility

```
##  Family: gaussian
##   Links: mu = identity
## Formula: om2 ~ diagnosis
##    Data: df.hgf (Number of observations: 66)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
##
## Regression Coefficients:
##            Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     -5.93      0.28    -6.48    -5.37 1.00     8326     6174
## diagnosis1     0.19      0.30    -0.39     0.78 1.00     7098     6186
## diagnosis2     0.15      0.30    -0.45     0.73 1.00     7272     5704
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     2.28      0.16     1.99     2.61 1.00     8008     5792
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

S3.2 Predicting ADHD diagnosis with HGF parameters

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: group ~ sbe1 + sbe2 + sbe3 + sze + som2 + som3
##    Data: df.hgf (Number of observations: 41)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
```

```
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     0.06      0.29    -0.52     0.63 1.00    11663     5569
## sbe1          0.18      0.36    -0.53     0.91 1.00    11572     5518
## sbe2         -0.09      0.37    -0.81     0.65 1.00    10220     6430
## sbe3         -0.19      0.36    -0.89     0.51 1.00    12722     6120
## sze           0.33      0.42    -0.48     1.17 1.00    10045     5970
## som2          0.66      0.36    -0.01     1.40 1.00    11259     6381
## som3         -0.04      0.44    -0.89     0.82 1.00     9145     6332
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

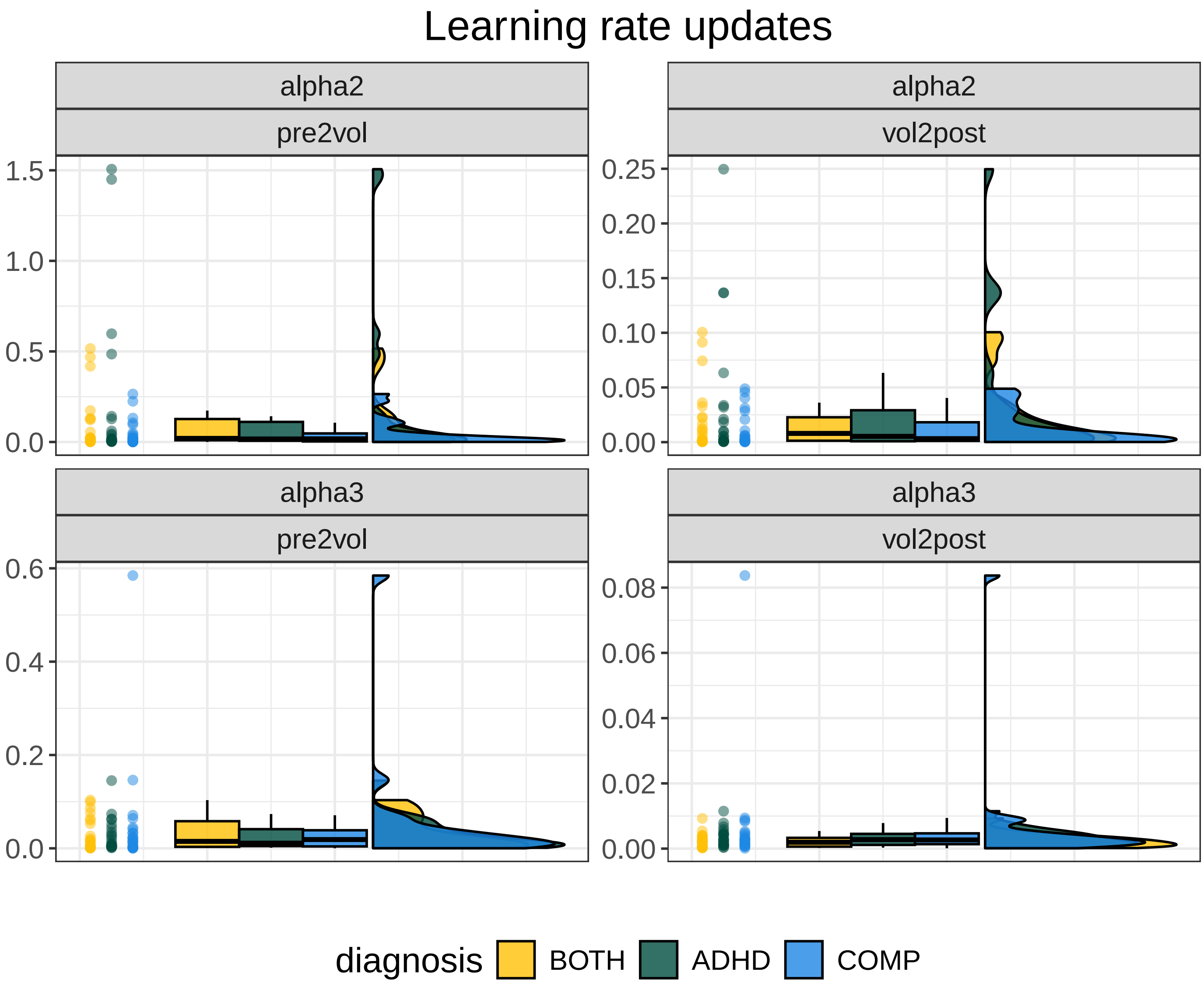
S3.3 Predicting ADHD medication with HGF parameters

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: group.meds ~ sbe1 + sbe2 + sbe3 + sze + som2 + som3
##    Data: df.hgf (Number of observations: 44)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     0.39      0.29    -0.17     0.96 1.00    13857     5726
## sbe1          0.19      0.35    -0.49     0.90 1.00    11114     6722
## sbe2          0.28      0.36    -0.43     1.00 1.00     9999     6075
## sbe3          0.04      0.33    -0.60     0.69 1.00    11532     6472
## sze           0.20      0.35    -0.46     0.91 1.00    11852     6181
## som2         -0.52      0.33    -1.20     0.11 1.00    10907     6430
## som3         -0.18      0.38    -0.94     0.56 1.00    11295     5957
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

S3.4 Learning rate update

```
##  Family: lognormal
##   Links: mu = identity
## Formula: value ~ diagnosis * level * change + (level + change | subID)
##    Data: df.upd (Number of observations: 264)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~subID (Number of levels: 66)
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(Intercept)                1.11      0.12     0.90     1.35 1.00     2373
## sd(level1)                   0.82      0.10     0.63     1.03 1.00     2986
## sd(change1)                  0.21      0.08     0.05     0.36 1.00     2793
## cor(Intercept,level1)        0.41      0.13     0.15     0.65 1.00     2399
## cor(Intercept,change1)       0.60      0.23     0.02     0.92 1.00     5210
## cor(level1,change1)          0.62      0.23     0.03     0.92 1.00     5018
##                          Tail_ESS
## sd(Intercept)                4179
## sd(level1)                   5020
## sd(change1)                  1653
```

```
## cor(Intercept,level1)      4229
## cor(Intercept,change1)     3847
## cor(level1,change1)        3687
##
## Regression Coefficients:
##                           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                    -4.95      0.15    -5.24    -4.66 1.00     1627
## diagnosis1                    0.18      0.21    -0.22     0.59 1.00     1712
## diagnosis2                   -0.07      0.21    -0.48     0.33 1.00     1640
## level1                        0.33      0.12     0.10     0.56 1.00     2420
## change1                       0.77      0.07     0.64     0.90 1.00     6477
## diagnosis1:level1             0.06      0.17    -0.26     0.39 1.00     2513
## diagnosis2:level1             0.15      0.16    -0.17     0.47 1.00     2632
## diagnosis1:change1            0.05      0.10    -0.14     0.24 1.00     5609
## diagnosis2:change1            0.04      0.10    -0.15     0.23 1.00     6322
## level1:change1               -0.10      0.06    -0.22     0.03 1.00    12332
## diagnosis1:level1:change1     0.06      0.09    -0.11     0.24 1.00     7030
## diagnosis2:level1:change1    -0.07      0.09    -0.24     0.10 1.00     7296
##                           Tail_ESS
## Intercept                     2760
## diagnosis1                    2534
## diagnosis2                    2841
## level1                        3706
## change1                       5573
## diagnosis1:level1             4210
## diagnosis2:level1             4199
## diagnosis1:change1            5829
## diagnosis2:change1            5894
## level1:change1                5555
## diagnosis1:level1:change1     6185
## diagnosis2:level1:change1     5620
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     1.01      0.07     0.89     1.15 1.00     2726     4036
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```



S3.5 Group differences in drift rate

```
##  Family: gaussian
##   Links: mu = identity
## Formula: v ~ diagnosis * phase + (1 | s)
##    Data: df.lng (Number of observations: 198)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~s (Number of levels: 66)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.59      0.06     0.49     0.72 1.00     1552     3009
##
## Regression Coefficients:
##                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept             2.31      0.08     2.16     2.47 1.01      912     1772
## diagnosis1           -0.27      0.11    -0.48    -0.07 1.01      883     1820
## diagnosis2           -0.05      0.11    -0.26     0.16 1.01      900     1974
## phase1               -0.03      0.03    -0.09     0.04 1.00     6249     5825
## phase2                0.04      0.03    -0.02     0.11 1.00     6359     5664
## diagnosis1:phase1     0.08      0.05    -0.01     0.17 1.00     5341     5891
## diagnosis2:phase1    -0.01      0.05    -0.11     0.08 1.00     5088     5845
## diagnosis1:phase2     0.01      0.05    -0.08     0.11 1.00     5421     5318
```

```
## diagnosis2:phase2    -0.03      0.05    -0.12     0.06 1.00     4940     5486
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.33      0.02     0.29     0.38 1.00     4744     5283
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

S3.6 Influence of medication on drift rate

```
##  Family: gaussian
##   Links: mu = identity
## Formula: v ~ adhd.meds.bin * phase + (1 | s)
##    Data: df.lng.sel (Number of observations: 132)
##   Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
##          total post-warmup draws = 8000
##
## Multilevel Hyperparameters:
## ~s (Number of levels: 44)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.55      0.07     0.44     0.70 1.00     1552     2718
##
## Regression Coefficients:
##                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                2.14      0.09     1.97     2.32 1.00     1049
## adhd.meds.bin1           0.06      0.09    -0.11     0.22 1.00     1057
## phase1                   0.01      0.03    -0.06     0.07 1.00     6948
## phase2                   0.03      0.03    -0.03     0.10 1.00     7233
## adhd.meds.bin1:phase1    0.01      0.03    -0.06     0.08 1.00     7207
## adhd.meds.bin1:phase2   -0.00      0.03    -0.07     0.06 1.00     7282
##                      Tail_ESS
## Intercept                2335
## adhd.meds.bin1           1984
## phase1                   6272
## phase2                   6017
## adhd.meds.bin1:phase1    5843
## adhd.meds.bin1:phase2    5958
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.28      0.02     0.24     0.32 1.00     4994     5476
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

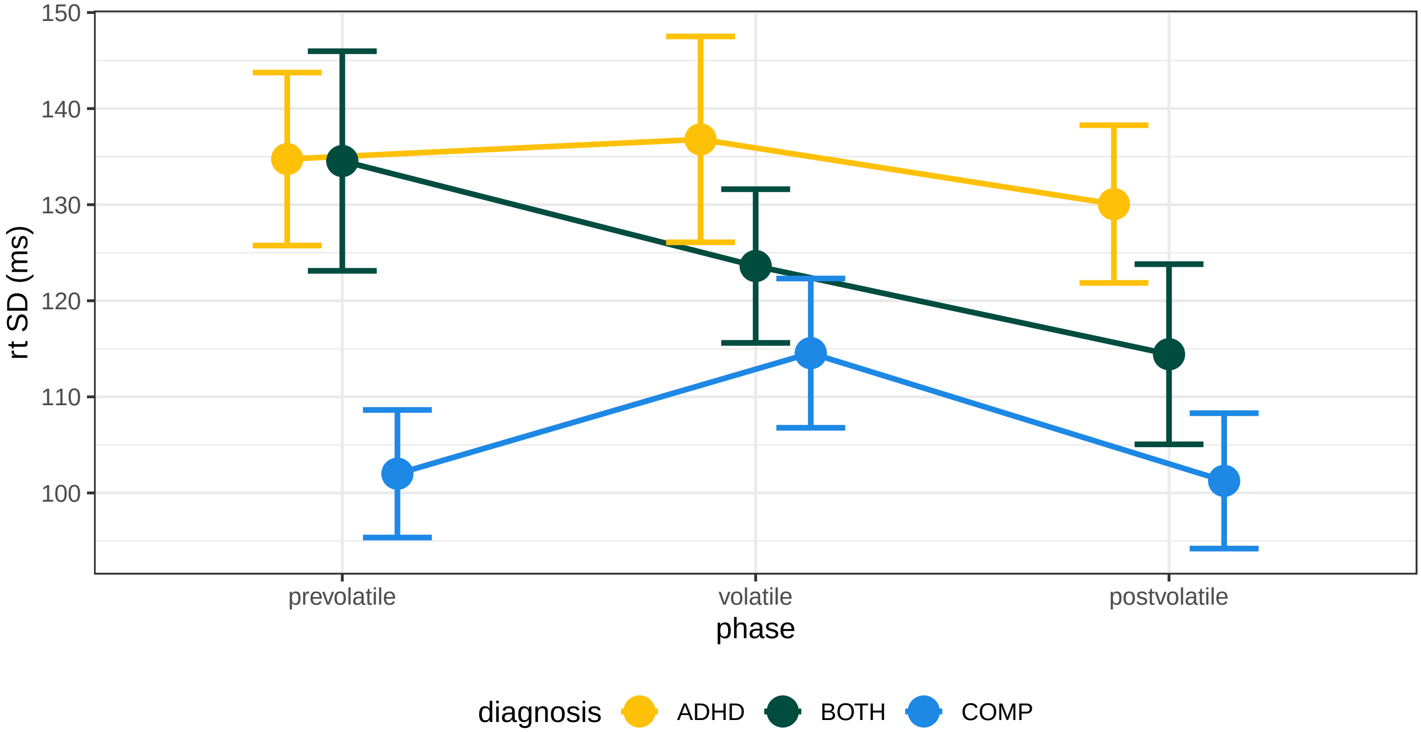
S4 Conventional analyses of responses and pupil sizes

S4.1 Response time variance

```
##  Family: lognormal
##   Links: mu = identity
## Formula: rt.var ~ diagnosis * expected * phase + (expected + phase | subID)
##    Data: df.var (Number of observations: 387)
##   Draws: 4 chains, each with iter = 6000; warmup = 1500; thin = 1;
##          total post-warmup draws = 18000
##
## Multilevel Hyperparameters:
```

```
## ~subID (Number of levels: 66)
##                           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(Intercept)                 0.36      0.03     0.30     0.43 1.00     3864
## sd(expected1)                 0.02      0.01     0.00     0.05 1.00     4508
## sd(phase1)                    0.09      0.02     0.05     0.12 1.00     6414
## sd(phase2)                    0.03      0.02     0.00     0.08 1.00     4058
## cor(Intercept,expected1)     -0.20      0.31    -0.74     0.47 1.00    22631
## cor(Intercept,phase1)        -0.23      0.18    -0.56     0.13 1.00    17311
## cor(expected1,phase1)         0.26      0.33    -0.48     0.81 1.00     2930
## cor(Intercept,phase2)         0.04      0.30    -0.56     0.62 1.00    22936
## cor(expected1,phase2)         0.19      0.37    -0.59     0.81 1.00     6950
## cor(phase1,phase2)            0.08      0.34    -0.58     0.70 1.00    17998
##                           Tail_ESS
## sd(Intercept)                 7484
## sd(expected1)                 5809
## sd(phase1)                    6234
## sd(phase2)                    6446
## cor(Intercept,expected1)     12093
## cor(Intercept,phase1)        13589
## cor(expected1,phase1)         5561
## cor(Intercept,phase2)        12392
## cor(expected1,phase2)        11325
## cor(phase1,phase2)           14323
##
## Regression Coefficients:
##                              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                        4.71      0.04     4.62     4.79 1.00     2209
## diagnosis1                       0.09      0.06    -0.03     0.21 1.00     2291
## diagnosis2                       0.03      0.06    -0.09     0.15 1.00     2444
## expected1                        0.06      0.01     0.04     0.08 1.00    30294
## phase1                           0.02      0.02    -0.01     0.05 1.00    15167
## phase2                           0.03      0.01     0.00     0.06 1.00    26066
## diagnosis1:expected1            -0.01      0.01    -0.04     0.02 1.00    22496
## diagnosis2:expected1            -0.00      0.01    -0.03     0.02 1.00    22150
## diagnosis1:phase1                0.00      0.02    -0.05     0.05 1.00    13212
## diagnosis2:phase1                0.05      0.02    -0.00     0.10 1.00    13450
## diagnosis1:phase2               -0.03      0.02    -0.07     0.01 1.00    19605
## diagnosis2:phase2               -0.02      0.02    -0.06     0.02 1.00    20126
## expected1:phase1                 0.02      0.01    -0.00     0.05 1.00    27027
## expected1:phase2                -0.02      0.01    -0.05     0.00 1.00    25645
## diagnosis1:expected1:phase1     -0.01      0.02    -0.05     0.03 1.00    19059
## diagnosis2:expected1:phase1     -0.01      0.02    -0.05     0.02 1.00    19306
## diagnosis1:expected1:phase2      0.04      0.02    -0.00     0.07 1.00    19797
## diagnosis2:expected1:phase2      0.00      0.02    -0.03     0.04 1.00    18557
##                              Tail_ESS
## Intercept                        5022
## diagnosis1                       4509
## diagnosis2                       4865
## expected1                       13918
## phase1                          13959
## phase2                          13147
## diagnosis1:expected1            14370
## diagnosis2:expected1            14145
## diagnosis1:phase1               13651
## diagnosis2:phase1               13448
## diagnosis1:phase2               13824
## diagnosis2:phase2               14268
## expected1:phase1                15127
## expected1:phase2                15707
## diagnosis1:expected1:phase1     15059
```

```
## diagnosis2:expected1:phase1    13226
## diagnosis1:expected1:phase2    14941
## diagnosis2:expected1:phase2    14865
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.18      0.01     0.17     0.20 1.00     6545    10033
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```



S4.2 Response times

```
##  Family: shifted_lognormal
##   Links: mu = identity
## Formula: rt.cor ~ diagnosis * expected * phase * difficulty + (expected * phase * difficulty | subID) + (di
##    Data: df.pal (Number of observations: 11207)
##   Draws: 4 chains, each with iter = 6000; warmup = 1500; thin = 1;
##          total post-warmup draws = 18000
##
## Multilevel Hyperparameters:
## ~subID (Number of levels: 44)
##                                                               Estimate
## sd(Intercept)                                                     0.16
## sd(expected1)                                                     0.01
## sd(phase1)                                                        0.06
## sd(phase2)                                                        0.02
## sd(difficulty1)                                                   0.01
## sd(difficulty2)                                                   0.01
## sd(expected1:phase1)                                              0.01
## sd(expected1:phase2)                                              0.01
## sd(expected1:difficulty1)                                         0.00
## sd(expected1:difficulty2)                                         0.01
## sd(phase1:difficulty1)                                            0.01
## sd(phase2:difficulty1)                                            0.01
```

```
## sd(phase1:difficulty2)                                           0.01
## sd(phase2:difficulty2)                                           0.01
## sd(expected1:phase1:difficulty1)                                 0.01
## sd(expected1:phase2:difficulty1)                                 0.01
## sd(expected1:phase1:difficulty2)                                 0.01
## sd(expected1:phase2:difficulty2)                                 0.01
## cor(Intercept,expected1)                                        -0.00
## cor(Intercept,phase1)                                           -0.04
## cor(expected1,phase1)                                            0.12
## cor(Intercept,phase2)                                           -0.01
## cor(expected1,phase2)                                           -0.01
## cor(phase1,phase2)                                              -0.14
## cor(Intercept,difficulty1)                                      -0.04
## cor(expected1,difficulty1)                                       0.07
## cor(phase1,difficulty1)                                         -0.16
## cor(phase2,difficulty1)                                          0.01
## cor(Intercept,difficulty2)                                      -0.02
## cor(expected1,difficulty2)                                       0.03
## cor(phase1,difficulty2)                                         -0.08
## cor(phase2,difficulty2)                                          0.01
## cor(difficulty1,difficulty2)                                     0.01
## cor(Intercept,expected1:phase1)                                 -0.04
## cor(expected1,expected1:phase1)                                  0.05
## cor(phase1,expected1:phase1)                                     0.04
## cor(phase2,expected1:phase1)                                    -0.08
## cor(difficulty1,expected1:phase1)                               -0.01
## cor(difficulty2,expected1:phase1)                                0.00
## cor(Intercept,expected1:phase2)                                  0.05
## cor(expected1,expected1:phase2)                                 -0.01
## cor(phase1,expected1:phase2)                                    -0.08
## cor(phase2,expected1:phase2)                                    -0.06
## cor(difficulty1,expected1:phase2)                                0.03
## cor(difficulty2,expected1:phase2)                                0.06
## cor(expected1:phase1,expected1:phase2)                          -0.04
## cor(Intercept,expected1:difficulty1)                            -0.04
## cor(expected1,expected1:difficulty1)                             0.02
## cor(phase1,expected1:difficulty1)                                0.01
## cor(phase2,expected1:difficulty1)                                0.05
## cor(difficulty1,expected1:difficulty1)                          -0.03
## cor(difficulty2,expected1:difficulty1)                          -0.00
## cor(expected1:phase1,expected1:difficulty1)                      0.00
## cor(expected1:phase2,expected1:difficulty1)                     -0.01
## cor(Intercept,expected1:difficulty2)                            -0.07
## cor(expected1,expected1:difficulty2)                             0.04
## cor(phase1,expected1:difficulty2)                                0.29
## cor(phase2,expected1:difficulty2)                                0.01
## cor(difficulty1,expected1:difficulty2)                          -0.09
## cor(difficulty2,expected1:difficulty2)                          -0.06
## cor(expected1:phase1,expected1:difficulty2)                     -0.00
## cor(expected1:phase2,expected1:difficulty2)                      0.04
## cor(expected1:difficulty1,expected1:difficulty2)                -0.04
## cor(Intercept,phase1:difficulty1)                               -0.02
## cor(expected1,phase1:difficulty1)                               -0.03
## cor(phase1,phase1:difficulty1)                                   0.02
## cor(phase2,phase1:difficulty1)                                   0.02
## cor(difficulty1,phase1:difficulty1)                             -0.01
## cor(difficulty2,phase1:difficulty1)                             -0.00
## cor(expected1:phase1,phase1:difficulty1)                        -0.02
## cor(expected1:phase2,phase1:difficulty1)                        -0.01
## cor(expected1:difficulty1,phase1:difficulty1)                    0.01
```

```
## cor(expected1:difficulty2,phase1:difficulty1)                      0.04
## cor(Intercept,phase2:difficulty1)                                 -0.03
## cor(expected1,phase2:difficulty1)                                  0.10
## cor(phase1,phase2:difficulty1)                                    -0.08
## cor(phase2,phase2:difficulty1)                                    -0.05
## cor(difficulty1,phase2:difficulty1)                                0.09
## cor(difficulty2,phase2:difficulty1)                                0.05
## cor(expected1:phase1,phase2:difficulty1)                          -0.02
## cor(expected1:phase2,phase2:difficulty1)                           0.02
## cor(expected1:difficulty1,phase2:difficulty1)                      0.02
## cor(expected1:difficulty2,phase2:difficulty1)                      0.02
## cor(phase1:difficulty1,phase2:difficulty1)                        -0.04
## cor(Intercept,phase1:difficulty2)                                  0.03
## cor(expected1,phase1:difficulty2)                                 -0.02
## cor(phase1,phase1:difficulty2)                                     0.04
## cor(phase2,phase1:difficulty2)                                     0.05
## cor(difficulty1,phase1:difficulty2)                               -0.04
## cor(difficulty2,phase1:difficulty2)                                0.00
## cor(expected1:phase1,phase1:difficulty2)                          -0.03
## cor(expected1:phase2,phase1:difficulty2)                          -0.01
## cor(expected1:difficulty1,phase1:difficulty2)                     -0.01
## cor(expected1:difficulty2,phase1:difficulty2)                      0.06
## cor(phase1:difficulty1,phase1:difficulty2)                        -0.02
## cor(phase2:difficulty1,phase1:difficulty2)                        -0.03
## cor(Intercept,phase2:difficulty2)                                  0.04
## cor(expected1,phase2:difficulty2)                                  0.01
## cor(phase1,phase2:difficulty2)                                    -0.08
## cor(phase2,phase2:difficulty2)                                    -0.01
## cor(difficulty1,phase2:difficulty2)                                0.02
## cor(difficulty2,phase2:difficulty2)                                0.01
## cor(expected1:phase1,phase2:difficulty2)                           0.02
## cor(expected1:phase2,phase2:difficulty2)                           0.02
## cor(expected1:difficulty1,phase2:difficulty2)                     -0.01
## cor(expected1:difficulty2,phase2:difficulty2)                     -0.04
## cor(phase1:difficulty1,phase2:difficulty2)                        -0.03
## cor(phase2:difficulty1,phase2:difficulty2)                        -0.05
## cor(phase1:difficulty2,phase2:difficulty2)                        -0.01
## cor(Intercept,expected1:phase1:difficulty1)                        0.04
## cor(expected1,expected1:phase1:difficulty1)                       -0.04
## cor(phase1,expected1:phase1:difficulty1)                          -0.01
## cor(phase2,expected1:phase1:difficulty1)                           0.05
## cor(difficulty1,expected1:phase1:difficulty1)                     -0.00
## cor(difficulty2,expected1:phase1:difficulty1)                     -0.01
## cor(expected1:phase1,expected1:phase1:difficulty1)                -0.01
## cor(expected1:phase2,expected1:phase1:difficulty1)                 0.01
## cor(expected1:difficulty1,expected1:phase1:difficulty1)            0.01
## cor(expected1:difficulty2,expected1:phase1:difficulty1)            0.00
## cor(phase1:difficulty1,expected1:phase1:difficulty1)              -0.02
## cor(phase2:difficulty1,expected1:phase1:difficulty1)              -0.03
## cor(phase1:difficulty2,expected1:phase1:difficulty1)              -0.02
## cor(phase2:difficulty2,expected1:phase1:difficulty1)              -0.02
## cor(Intercept,expected1:phase2:difficulty1)                        0.10
## cor(expected1,expected1:phase2:difficulty1)                        0.02
## cor(phase1,expected1:phase2:difficulty1)                           0.04
## cor(phase2,expected1:phase2:difficulty1)                           0.07
## cor(difficulty1,expected1:phase2:difficulty1)                     -0.00
## cor(difficulty2,expected1:phase2:difficulty1)                      0.00
## cor(expected1:phase1,expected1:phase2:difficulty1)                -0.00
## cor(expected1:phase2,expected1:phase2:difficulty1)                 0.01
## cor(expected1:difficulty1,expected1:phase2:difficulty1)           -0.00
```

```
## cor(expected1:difficulty2,expected1:phase2:difficulty1)            0.00
## cor(phase1:difficulty1,expected1:phase2:difficulty1)              -0.01
## cor(phase2:difficulty1,expected1:phase2:difficulty1)              -0.04
## cor(phase1:difficulty2,expected1:phase2:difficulty1)               0.00
## cor(phase2:difficulty2,expected1:phase2:difficulty1)              -0.02
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1)    -0.01
## cor(Intercept,expected1:phase1:difficulty2)                        0.04
## cor(expected1,expected1:phase1:difficulty2)                       -0.06
## cor(phase1,expected1:phase1:difficulty2)                          -0.06
## cor(phase2,expected1:phase1:difficulty2)                           0.10
## cor(difficulty1,expected1:phase1:difficulty2)                     -0.01
## cor(difficulty2,expected1:phase1:difficulty2)                      0.03
## cor(expected1:phase1,expected1:phase1:difficulty2)                -0.05
## cor(expected1:phase2,expected1:phase1:difficulty2)                 0.03
## cor(expected1:difficulty1,expected1:phase1:difficulty2)           -0.01
## cor(expected1:difficulty2,expected1:phase1:difficulty2)            0.04
## cor(phase1:difficulty1,expected1:phase1:difficulty2)              -0.01
## cor(phase2:difficulty1,expected1:phase1:difficulty2)              -0.02
## cor(phase1:difficulty2,expected1:phase1:difficulty2)              -0.02
## cor(phase2:difficulty2,expected1:phase1:difficulty2)              -0.00
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2)    -0.01
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2)     0.01
## cor(Intercept,expected1:phase2:difficulty2)                        0.09
## cor(expected1,expected1:phase2:difficulty2)                       -0.01
## cor(phase1,expected1:phase2:difficulty2)                          -0.04
## cor(phase2,expected1:phase2:difficulty2)                           0.08
## cor(difficulty1,expected1:phase2:difficulty2)                      0.00
## cor(difficulty2,expected1:phase2:difficulty2)                      0.01
## cor(expected1:phase1,expected1:phase2:difficulty2)                 0.00
## cor(expected1:phase2,expected1:phase2:difficulty2)                 0.03
## cor(expected1:difficulty1,expected1:phase2:difficulty2)           -0.02
## cor(expected1:difficulty2,expected1:phase2:difficulty2)           -0.03
## cor(phase1:difficulty1,expected1:phase2:difficulty2)              -0.00
## cor(phase2:difficulty1,expected1:phase2:difficulty2)              -0.04
## cor(phase1:difficulty2,expected1:phase2:difficulty2)               0.00
## cor(phase2:difficulty2,expected1:phase2:difficulty2)              -0.02
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2)     0.01
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2)    -0.02
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2)     0.01
##                                                                 Est.Error
## sd(Intercept)                                                        0.02
## sd(expected1)                                                        0.01
## sd(phase1)                                                           0.01
## sd(phase2)                                                           0.00
## sd(difficulty1)                                                      0.00
## sd(difficulty2)                                                      0.00
## sd(expected1:phase1)                                                 0.01
## sd(expected1:phase2)                                                 0.00
## sd(expected1:difficulty1)                                            0.00
## sd(expected1:difficulty2)                                            0.00
## sd(phase1:difficulty1)                                               0.00
## sd(phase2:difficulty1)                                               0.01
## sd(phase1:difficulty2)                                               0.00
## sd(phase2:difficulty2)                                               0.01
## sd(expected1:phase1:difficulty1)                                     0.00
## sd(expected1:phase2:difficulty1)                                     0.00
## sd(expected1:phase1:difficulty2)                                     0.01
## sd(expected1:phase2:difficulty2)                                     0.00
## cor(Intercept,expected1)                                             0.18
## cor(Intercept,phase1)                                                0.13
```

```
## cor(expected1,phase1)                                          0.18
## cor(Intercept,phase2)                                          0.15
## cor(expected1,phase2)                                          0.19
## cor(phase1,phase2)                                             0.16
## cor(Intercept,difficulty1)                                     0.20
## cor(expected1,difficulty1)                                     0.21
## cor(phase1,difficulty1)                                        0.21
## cor(phase2,difficulty1)                                        0.21
## cor(Intercept,difficulty2)                                     0.21
## cor(expected1,difficulty2)                                     0.21
## cor(phase1,difficulty2)                                        0.21
## cor(phase2,difficulty2)                                        0.21
## cor(difficulty1,difficulty2)                                   0.22
## cor(Intercept,expected1:phase1)                                0.20
## cor(expected1,expected1:phase1)                                0.22
## cor(phase1,expected1:phase1)                                   0.20
## cor(phase2,expected1:phase1)                                   0.21
## cor(difficulty1,expected1:phase1)                              0.22
## cor(difficulty2,expected1:phase1)                              0.22
## cor(Intercept,expected1:phase2)                                0.21
## cor(expected1,expected1:phase2)                                0.22
## cor(phase1,expected1:phase2)                                   0.21
## cor(phase2,expected1:phase2)                                   0.22
## cor(difficulty1,expected1:phase2)                              0.22
## cor(difficulty2,expected1:phase2)                              0.22
## cor(expected1:phase1,expected1:phase2)                         0.22
## cor(Intercept,expected1:difficulty1)                           0.21
## cor(expected1,expected1:difficulty1)                           0.22
## cor(phase1,expected1:difficulty1)                              0.21
## cor(phase2,expected1:difficulty1)                              0.21
## cor(difficulty1,expected1:difficulty1)                         0.22
## cor(difficulty2,expected1:difficulty1)                         0.22
## cor(expected1:phase1,expected1:difficulty1)                    0.22
## cor(expected1:phase2,expected1:difficulty1)                    0.22
## cor(Intercept,expected1:difficulty2)                           0.17
## cor(expected1,expected1:difficulty2)                           0.20
## cor(phase1,expected1:difficulty2)                              0.17
## cor(phase2,expected1:difficulty2)                              0.19
## cor(difficulty1,expected1:difficulty2)                         0.22
## cor(difficulty2,expected1:difficulty2)                         0.22
## cor(expected1:phase1,expected1:difficulty2)                    0.21
## cor(expected1:phase2,expected1:difficulty2)                    0.22
## cor(expected1:difficulty1,expected1:difficulty2)               0.22
## cor(Intercept,phase1:difficulty1)                              0.21
## cor(expected1,phase1:difficulty1)                              0.22
## cor(phase1,phase1:difficulty1)                                 0.21
## cor(phase2,phase1:difficulty1)                                 0.22
## cor(difficulty1,phase1:difficulty1)                            0.22
## cor(difficulty2,phase1:difficulty1)                            0.22
## cor(expected1:phase1,phase1:difficulty1)                       0.22
## cor(expected1:phase2,phase1:difficulty1)                       0.22
## cor(expected1:difficulty1,phase1:difficulty1)                  0.22
## cor(expected1:difficulty2,phase1:difficulty1)                  0.22
## cor(Intercept,phase2:difficulty1)                              0.19
## cor(expected1,phase2:difficulty1)                              0.20
## cor(phase1,phase2:difficulty1)                                 0.19
## cor(phase2,phase2:difficulty1)                                 0.20
## cor(difficulty1,phase2:difficulty1)                            0.21
## cor(difficulty2,phase2:difficulty1)                            0.22
## cor(expected1:phase1,phase2:difficulty1)                       0.22
```

```
## cor(expected1:phase2,phase2:difficulty1)                          0.22
## cor(expected1:difficulty1,phase2:difficulty1)                     0.22
## cor(expected1:difficulty2,phase2:difficulty1)                     0.20
## cor(phase1:difficulty1,phase2:difficulty1)                        0.22
## cor(Intercept,phase1:difficulty2)                                 0.21
## cor(expected1,phase1:difficulty2)                                 0.22
## cor(phase1,phase1:difficulty2)                                    0.22
## cor(phase2,phase1:difficulty2)                                    0.22
## cor(difficulty1,phase1:difficulty2)                               0.22
## cor(difficulty2,phase1:difficulty2)                               0.22
## cor(expected1:phase1,phase1:difficulty2)                          0.22
## cor(expected1:phase2,phase1:difficulty2)                          0.22
## cor(expected1:difficulty1,phase1:difficulty2)                     0.22
## cor(expected1:difficulty2,phase1:difficulty2)                     0.21
## cor(phase1:difficulty1,phase1:difficulty2)                        0.22
## cor(phase2:difficulty1,phase1:difficulty2)                        0.22
## cor(Intercept,phase2:difficulty2)                                 0.20
## cor(expected1,phase2:difficulty2)                                 0.21
## cor(phase1,phase2:difficulty2)                                    0.20
## cor(phase2,phase2:difficulty2)                                    0.21
## cor(difficulty1,phase2:difficulty2)                               0.21
## cor(difficulty2,phase2:difficulty2)                               0.22
## cor(expected1:phase1,phase2:difficulty2)                          0.22
## cor(expected1:phase2,phase2:difficulty2)                          0.22
## cor(expected1:difficulty1,phase2:difficulty2)                     0.22
## cor(expected1:difficulty2,phase2:difficulty2)                     0.21
## cor(phase1:difficulty1,phase2:difficulty2)                        0.22
## cor(phase2:difficulty1,phase2:difficulty2)                        0.22
## cor(phase1:difficulty2,phase2:difficulty2)                        0.22
## cor(Intercept,expected1:phase1:difficulty1)                       0.21
## cor(expected1,expected1:phase1:difficulty1)                       0.21
## cor(phase1,expected1:phase1:difficulty1)                          0.21
## cor(phase2,expected1:phase1:difficulty1)                          0.22
## cor(difficulty1,expected1:phase1:difficulty1)                     0.22
## cor(difficulty2,expected1:phase1:difficulty1)                     0.22
## cor(expected1:phase1,expected1:phase1:difficulty1)                0.22
## cor(expected1:phase2,expected1:phase1:difficulty1)                0.22
## cor(expected1:difficulty1,expected1:phase1:difficulty1)           0.22
## cor(expected1:difficulty2,expected1:phase1:difficulty1)           0.21
## cor(phase1:difficulty1,expected1:phase1:difficulty1)              0.22
## cor(phase2:difficulty1,expected1:phase1:difficulty1)              0.22
## cor(phase1:difficulty2,expected1:phase1:difficulty1)              0.22
## cor(phase2:difficulty2,expected1:phase1:difficulty1)              0.22
## cor(Intercept,expected1:phase2:difficulty1)                       0.21
## cor(expected1,expected1:phase2:difficulty1)                       0.22
## cor(phase1,expected1:phase2:difficulty1)                          0.21
## cor(phase2,expected1:phase2:difficulty1)                          0.22
## cor(difficulty1,expected1:phase2:difficulty1)                     0.22
## cor(difficulty2,expected1:phase2:difficulty1)                     0.22
## cor(expected1:phase1,expected1:phase2:difficulty1)                0.22
## cor(expected1:phase2,expected1:phase2:difficulty1)                0.22
## cor(expected1:difficulty1,expected1:phase2:difficulty1)           0.22
## cor(expected1:difficulty2,expected1:phase2:difficulty1)           0.21
## cor(phase1:difficulty1,expected1:phase2:difficulty1)              0.22
## cor(phase2:difficulty1,expected1:phase2:difficulty1)              0.22
## cor(phase1:difficulty2,expected1:phase2:difficulty1)              0.22
## cor(phase2:difficulty2,expected1:phase2:difficulty1)              0.22
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1)    0.22
## cor(Intercept,expected1:phase1:difficulty2)                       0.21
## cor(expected1,expected1:phase1:difficulty2)                       0.22
```

```
## cor(phase1,expected1:phase1:difficulty2)                          0.21
## cor(phase2,expected1:phase1:difficulty2)                          0.21
## cor(difficulty1,expected1:phase1:difficulty2)                     0.22
## cor(difficulty2,expected1:phase1:difficulty2)                     0.22
## cor(expected1:phase1,expected1:phase1:difficulty2)                0.22
## cor(expected1:phase2,expected1:phase1:difficulty2)                0.22
## cor(expected1:difficulty1,expected1:phase1:difficulty2)           0.22
## cor(expected1:difficulty2,expected1:phase1:difficulty2)           0.21
## cor(phase1:difficulty1,expected1:phase1:difficulty2)              0.22
## cor(phase2:difficulty1,expected1:phase1:difficulty2)              0.22
## cor(phase1:difficulty2,expected1:phase1:difficulty2)              0.22
## cor(phase2:difficulty2,expected1:phase1:difficulty2)              0.22
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2)    0.22
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2)    0.22
## cor(Intercept,expected1:phase2:difficulty2)                       0.22
## cor(expected1,expected1:phase2:difficulty2)                       0.22
## cor(phase1,expected1:phase2:difficulty2)                          0.21
## cor(phase2,expected1:phase2:difficulty2)                          0.22
## cor(difficulty1,expected1:phase2:difficulty2)                     0.22
## cor(difficulty2,expected1:phase2:difficulty2)                     0.22
## cor(expected1:phase1,expected1:phase2:difficulty2)                0.22
## cor(expected1:phase2,expected1:phase2:difficulty2)                0.22
## cor(expected1:difficulty1,expected1:phase2:difficulty2)           0.22
## cor(expected1:difficulty2,expected1:phase2:difficulty2)           0.22
## cor(phase1:difficulty1,expected1:phase2:difficulty2)              0.22
## cor(phase2:difficulty1,expected1:phase2:difficulty2)              0.22
## cor(phase1:difficulty2,expected1:phase2:difficulty2)              0.22
## cor(phase2:difficulty2,expected1:phase2:difficulty2)              0.22
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2)    0.22
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2)    0.22
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2)    0.22
##                                                                 l-95% CI
## sd(Intercept)                                                       0.13
## sd(expected1)                                                       0.00
## sd(phase1)                                                          0.04
## sd(phase2)                                                          0.01
## sd(difficulty1)                                                     0.00
## sd(difficulty2)                                                     0.00
## sd(expected1:phase1)                                                0.00
## sd(expected1:phase2)                                                0.00
## sd(expected1:difficulty1)                                           0.00
## sd(expected1:difficulty2)                                           0.00
## sd(phase1:difficulty1)                                              0.00
## sd(phase2:difficulty1)                                              0.00
## sd(phase1:difficulty2)                                              0.00
## sd(phase2:difficulty2)                                              0.00
## sd(expected1:phase1:difficulty1)                                    0.00
## sd(expected1:phase2:difficulty1)                                    0.00
## sd(expected1:phase1:difficulty2)                                    0.00
## sd(expected1:phase2:difficulty2)                                    0.00
## cor(Intercept,expected1)                                           -0.34
## cor(Intercept,phase1)                                              -0.30
## cor(expected1,phase1)                                              -0.24
## cor(Intercept,phase2)                                              -0.31
## cor(expected1,phase2)                                              -0.39
## cor(phase1,phase2)                                                 -0.45
## cor(Intercept,difficulty1)                                         -0.42
## cor(expected1,difficulty1)                                         -0.36
## cor(phase1,difficulty1)                                            -0.54
## cor(phase2,difficulty1)                                            -0.40
```

```
## cor(Intercept,difficulty2)                                  -0.42
## cor(expected1,difficulty2)                                  -0.38
## cor(phase1,difficulty2)                                     -0.48
## cor(phase2,difficulty2)                                     -0.40
## cor(difficulty1,difficulty2)                                -0.41
## cor(Intercept,expected1:phase1)                             -0.42
## cor(expected1,expected1:phase1)                             -0.38
## cor(phase1,expected1:phase1)                                -0.36
## cor(phase2,expected1:phase1)                                -0.47
## cor(difficulty1,expected1:phase1)                           -0.42
## cor(difficulty2,expected1:phase1)                           -0.42
## cor(Intercept,expected1:phase2)                             -0.36
## cor(expected1,expected1:phase2)                             -0.43
## cor(phase1,expected1:phase2)                                -0.48
## cor(phase2,expected1:phase2)                                -0.48
## cor(difficulty1,expected1:phase2)                           -0.39
## cor(difficulty2,expected1:phase2)                           -0.38
## cor(expected1:phase1,expected1:phase2)                      -0.47
## cor(Intercept,expected1:difficulty1)                        -0.44
## cor(expected1,expected1:difficulty1)                        -0.41
## cor(phase1,expected1:difficulty1)                           -0.41
## cor(phase2,expected1:difficulty1)                           -0.37
## cor(difficulty1,expected1:difficulty1)                      -0.46
## cor(difficulty2,expected1:difficulty1)                      -0.43
## cor(expected1:phase1,expected1:difficulty1)                 -0.42
## cor(expected1:phase2,expected1:difficulty1)                 -0.43
## cor(Intercept,expected1:difficulty2)                        -0.40
## cor(expected1,expected1:difficulty2)                        -0.35
## cor(phase1,expected1:difficulty2)                           -0.08
## cor(phase2,expected1:difficulty2)                           -0.36
## cor(difficulty1,expected1:difficulty2)                      -0.49
## cor(difficulty2,expected1:difficulty2)                      -0.47
## cor(expected1:phase1,expected1:difficulty2)                 -0.41
## cor(expected1:phase2,expected1:difficulty2)                 -0.38
## cor(expected1:difficulty1,expected1:difficulty2)            -0.46
## cor(Intercept,phase1:difficulty1)                           -0.43
## cor(expected1,phase1:difficulty1)                           -0.45
## cor(phase1,phase1:difficulty1)                              -0.40
## cor(phase2,phase1:difficulty1)                              -0.41
## cor(difficulty1,phase1:difficulty1)                         -0.43
## cor(difficulty2,phase1:difficulty1)                         -0.43
## cor(expected1:phase1,phase1:difficulty1)                    -0.44
## cor(expected1:phase2,phase1:difficulty1)                    -0.43
## cor(expected1:difficulty1,phase1:difficulty1)               -0.42
## cor(expected1:difficulty2,phase1:difficulty1)               -0.39
## cor(Intercept,phase2:difficulty1)                           -0.39
## cor(expected1,phase2:difficulty1)                           -0.31
## cor(phase1,phase2:difficulty1)                              -0.43
## cor(phase2,phase2:difficulty1)                              -0.43
## cor(difficulty1,phase2:difficulty1)                         -0.33
## cor(difficulty2,phase2:difficulty1)                         -0.38
## cor(expected1:phase1,phase2:difficulty1)                    -0.43
## cor(expected1:phase2,phase2:difficulty1)                    -0.40
## cor(expected1:difficulty1,phase2:difficulty1)               -0.40
## cor(expected1:difficulty2,phase2:difficulty1)               -0.38
## cor(phase1:difficulty1,phase2:difficulty1)                  -0.46
## cor(Intercept,phase1:difficulty2)                           -0.39
## cor(expected1,phase1:difficulty2)                           -0.44
## cor(phase1,phase1:difficulty2)                              -0.38
## cor(phase2,phase1:difficulty2)                              -0.38
```

```
## cor(difficulty1,phase1:difficulty2)                           -0.45
## cor(difficulty2,phase1:difficulty2)                           -0.42
## cor(expected1:phase1,phase1:difficulty2)                      -0.45
## cor(expected1:phase2,phase1:difficulty2)                      -0.43
## cor(expected1:difficulty1,phase1:difficulty2)                 -0.43
## cor(expected1:difficulty2,phase1:difficulty2)                 -0.36
## cor(phase1:difficulty1,phase1:difficulty2)                    -0.44
## cor(phase2:difficulty1,phase1:difficulty2)                    -0.45
## cor(Intercept,phase2:difficulty2)                             -0.35
## cor(expected1,phase2:difficulty2)                             -0.41
## cor(phase1,phase2:difficulty2)                                -0.47
## cor(phase2,phase2:difficulty2)                                -0.40
## cor(difficulty1,phase2:difficulty2)                           -0.39
## cor(difficulty2,phase2:difficulty2)                           -0.41
## cor(expected1:phase1,phase2:difficulty2)                      -0.41
## cor(expected1:phase2,phase2:difficulty2)                      -0.40
## cor(expected1:difficulty1,phase2:difficulty2)                 -0.44
## cor(expected1:difficulty2,phase2:difficulty2)                 -0.44
## cor(phase1:difficulty1,phase2:difficulty2)                    -0.45
## cor(phase2:difficulty1,phase2:difficulty2)                    -0.46
## cor(phase1:difficulty2,phase2:difficulty2)                    -0.43
## cor(Intercept,expected1:phase1:difficulty1)                   -0.37
## cor(expected1,expected1:phase1:difficulty1)                   -0.45
## cor(phase1,expected1:phase1:difficulty1)                      -0.42
## cor(phase2,expected1:phase1:difficulty1)                      -0.38
## cor(difficulty1,expected1:phase1:difficulty1)                 -0.42
## cor(difficulty2,expected1:phase1:difficulty1)                 -0.44
## cor(expected1:phase1,expected1:phase1:difficulty1)            -0.43
## cor(expected1:phase2,expected1:phase1:difficulty1)            -0.42
## cor(expected1:difficulty1,expected1:phase1:difficulty1)       -0.41
## cor(expected1:difficulty2,expected1:phase1:difficulty1)       -0.41
## cor(phase1:difficulty1,expected1:phase1:difficulty1)          -0.45
## cor(phase2:difficulty1,expected1:phase1:difficulty1)          -0.45
## cor(phase1:difficulty2,expected1:phase1:difficulty1)          -0.44
## cor(phase2:difficulty2,expected1:phase1:difficulty1)          -0.44
## cor(Intercept,expected1:phase2:difficulty1)                   -0.33
## cor(expected1,expected1:phase2:difficulty1)                   -0.40
## cor(phase1,expected1:phase2:difficulty1)                      -0.38
## cor(phase2,expected1:phase2:difficulty1)                      -0.35
## cor(difficulty1,expected1:phase2:difficulty1)                 -0.42
## cor(difficulty2,expected1:phase2:difficulty1)                 -0.41
## cor(expected1:phase1,expected1:phase2:difficulty1)            -0.42
## cor(expected1:phase2,expected1:phase2:difficulty1)            -0.42
## cor(expected1:difficulty1,expected1:phase2:difficulty1)       -0.43
## cor(expected1:difficulty2,expected1:phase2:difficulty1)       -0.41
## cor(phase1:difficulty1,expected1:phase2:difficulty1)          -0.43
## cor(phase2:difficulty1,expected1:phase2:difficulty1)          -0.45
## cor(phase1:difficulty2,expected1:phase2:difficulty1)          -0.42
## cor(phase2:difficulty2,expected1:phase2:difficulty1)          -0.44
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1) -0.43
## cor(Intercept,expected1:phase1:difficulty2)                   -0.37
## cor(expected1,expected1:phase1:difficulty2)                   -0.47
## cor(phase1,expected1:phase1:difficulty2)                      -0.44
## cor(phase2,expected1:phase1:difficulty2)                      -0.33
## cor(difficulty1,expected1:phase1:difficulty2)                 -0.43
## cor(difficulty2,expected1:phase1:difficulty2)                 -0.40
## cor(expected1:phase1,expected1:phase1:difficulty2)            -0.46
## cor(expected1:phase2,expected1:phase1:difficulty2)            -0.39
## cor(expected1:difficulty1,expected1:phase1:difficulty2)       -0.43
## cor(expected1:difficulty2,expected1:phase1:difficulty2)       -0.38
```

```
## cor(phase1:difficulty1,expected1:phase1:difficulty2)             -0.43
## cor(phase2:difficulty1,expected1:phase1:difficulty2)             -0.43
## cor(phase1:difficulty2,expected1:phase1:difficulty2)             -0.44
## cor(phase2:difficulty2,expected1:phase1:difficulty2)             -0.42
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2)    -0.43
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2)    -0.41
## cor(Intercept,expected1:phase2:difficulty2)                       -0.34
## cor(expected1,expected1:phase2:difficulty2)                       -0.43
## cor(phase1,expected1:phase2:difficulty2)                          -0.45
## cor(phase2,expected1:phase2:difficulty2)                          -0.35
## cor(difficulty1,expected1:phase2:difficulty2)                     -0.42
## cor(difficulty2,expected1:phase2:difficulty2)                     -0.41
## cor(expected1:phase1,expected1:phase2:difficulty2)                -0.42
## cor(expected1:phase2,expected1:phase2:difficulty2)                -0.40
## cor(expected1:difficulty1,expected1:phase2:difficulty2)           -0.44
## cor(expected1:difficulty2,expected1:phase2:difficulty2)           -0.45
## cor(phase1:difficulty1,expected1:phase2:difficulty2)              -0.42
## cor(phase2:difficulty1,expected1:phase2:difficulty2)              -0.45
## cor(phase1:difficulty2,expected1:phase2:difficulty2)              -0.41
## cor(phase2:difficulty2,expected1:phase2:difficulty2)              -0.43
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2)    -0.41
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2)    -0.44
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2)    -0.42
##                                                                 u-95% CI Rhat
## sd(Intercept)                                                       0.20 1.00
## sd(expected1)                                                       0.02 1.00
## sd(phase1)                                                          0.07 1.00
## sd(phase2)                                                          0.03 1.00
## sd(difficulty1)                                                     0.02 1.00
## sd(difficulty2)                                                     0.02 1.00
## sd(expected1:phase1)                                                0.02 1.00
## sd(expected1:phase2)                                                0.02 1.00
## sd(expected1:difficulty1)                                           0.01 1.00
## sd(expected1:difficulty2)                                           0.02 1.00
## sd(phase1:difficulty1)                                              0.02 1.00
## sd(phase2:difficulty1)                                              0.03 1.00
## sd(phase1:difficulty2)                                              0.02 1.00
## sd(phase2:difficulty2)                                              0.02 1.00
## sd(expected1:phase1:difficulty1)                                    0.02 1.00
## sd(expected1:phase2:difficulty1)                                    0.02 1.00
## sd(expected1:phase1:difficulty2)                                    0.02 1.00
## sd(expected1:phase2:difficulty2)                                    0.02 1.00
## cor(Intercept,expected1)                                            0.35 1.00
## cor(Intercept,phase1)                                               0.22 1.00
## cor(expected1,phase1)                                               0.46 1.00
## cor(Intercept,phase2)                                               0.29 1.00
## cor(expected1,phase2)                                               0.37 1.00
## cor(phase1,phase2)                                                  0.18 1.00
## cor(Intercept,difficulty1)                                          0.35 1.00
## cor(expected1,difficulty1)                                          0.46 1.00
## cor(phase1,difficulty1)                                             0.26 1.00
## cor(phase2,difficulty1)                                             0.41 1.00
## cor(Intercept,difficulty2)                                          0.39 1.00
## cor(expected1,difficulty2)                                          0.44 1.00
## cor(phase1,difficulty2)                                             0.34 1.00
## cor(phase2,difficulty2)                                             0.42 1.00
## cor(difficulty1,difficulty2)                                        0.43 1.00
## cor(Intercept,expected1:phase1)                                     0.36 1.00
## cor(expected1,expected1:phase1)                                     0.47 1.00
## cor(phase1,expected1:phase1)                                        0.43 1.00
```

```
## cor(phase2,expected1:phase1)                                          0.35 1.00
## cor(difficulty1,expected1:phase1)                                     0.41 1.00
## cor(difficulty2,expected1:phase1)                                     0.42 1.00
## cor(Intercept,expected1:phase2)                                       0.45 1.00
## cor(expected1,expected1:phase2)                                       0.42 1.00
## cor(phase1,expected1:phase2)                                          0.34 1.00
## cor(phase2,expected1:phase2)                                          0.36 1.00
## cor(difficulty1,expected1:phase2)                                     0.45 1.00
## cor(difficulty2,expected1:phase2)                                     0.48 1.00
## cor(expected1:phase1,expected1:phase2)                                0.38 1.00
## cor(Intercept,expected1:difficulty1)                                  0.38 1.00
## cor(expected1,expected1:difficulty1)                                  0.44 1.00
## cor(phase1,expected1:difficulty1)                                     0.41 1.00
## cor(phase2,expected1:difficulty1)                                     0.46 1.00
## cor(difficulty1,expected1:difficulty1)                                0.41 1.00
## cor(difficulty2,expected1:difficulty1)                                0.42 1.00
## cor(expected1:phase1,expected1:difficulty1)                           0.42 1.00
## cor(expected1:phase2,expected1:difficulty1)                           0.41 1.00
## cor(Intercept,expected1:difficulty2)                                  0.27 1.00
## cor(expected1,expected1:difficulty2)                                  0.43 1.00
## cor(phase1,expected1:difficulty2)                                     0.60 1.00
## cor(phase2,expected1:difficulty2)                                     0.38 1.00
## cor(difficulty1,expected1:difficulty2)                                0.34 1.00
## cor(difficulty2,expected1:difficulty2)                                0.37 1.00
## cor(expected1:phase1,expected1:difficulty2)                           0.41 1.00
## cor(expected1:phase2,expected1:difficulty2)                           0.45 1.00
## cor(expected1:difficulty1,expected1:difficulty2)                      0.39 1.00
## cor(Intercept,phase1:difficulty1)                                     0.39 1.00
## cor(expected1,phase1:difficulty1)                                     0.39 1.00
## cor(phase1,phase1:difficulty1)                                        0.43 1.00
## cor(phase2,phase1:difficulty1)                                        0.43 1.00
## cor(difficulty1,phase1:difficulty1)                                   0.42 1.00
## cor(difficulty2,phase1:difficulty1)                                   0.42 1.00
## cor(expected1:phase1,phase1:difficulty1)                              0.40 1.00
## cor(expected1:phase2,phase1:difficulty1)                              0.41 1.00
## cor(expected1:difficulty1,phase1:difficulty1)                         0.43 1.00
## cor(expected1:difficulty2,phase1:difficulty1)                         0.45 1.00
## cor(Intercept,phase2:difficulty1)                                     0.33 1.00
## cor(expected1,phase2:difficulty1)                                     0.48 1.00
## cor(phase1,phase2:difficulty1)                                        0.29 1.00
## cor(phase2,phase2:difficulty1)                                        0.34 1.00
## cor(difficulty1,phase2:difficulty1)                                   0.49 1.00
## cor(difficulty2,phase2:difficulty1)                                   0.46 1.00
## cor(expected1:phase1,phase2:difficulty1)                              0.40 1.00
## cor(expected1:phase2,phase2:difficulty1)                              0.44 1.00
## cor(expected1:difficulty1,phase2:difficulty1)                         0.44 1.00
## cor(expected1:difficulty2,phase2:difficulty1)                         0.42 1.00
## cor(phase1:difficulty1,phase2:difficulty1)                            0.39 1.00
## cor(Intercept,phase1:difficulty2)                                     0.44 1.00
## cor(expected1,phase1:difficulty2)                                     0.40 1.00
## cor(phase1,phase1:difficulty2)                                        0.45 1.00
## cor(phase2,phase1:difficulty2)                                        0.47 1.00
## cor(difficulty1,phase1:difficulty2)                                   0.39 1.00
## cor(difficulty2,phase1:difficulty2)                                   0.42 1.00
## cor(expected1:phase1,phase1:difficulty2)                              0.40 1.00
## cor(expected1:phase2,phase1:difficulty2)                              0.41 1.00
## cor(expected1:difficulty1,phase1:difficulty2)                         0.42 1.00
## cor(expected1:difficulty2,phase1:difficulty2)                         0.46 1.00
## cor(phase1:difficulty1,phase1:difficulty2)                            0.42 1.00
## cor(phase2:difficulty1,phase1:difficulty2)                            0.39 1.00
```

```
## cor(Intercept,phase2:difficulty2)                                   0.42 1.00
## cor(expected1,phase2:difficulty2)                                   0.42 1.00
## cor(phase1,phase2:difficulty2)                                      0.33 1.00
## cor(phase2,phase2:difficulty2)                                      0.40 1.00
## cor(difficulty1,phase2:difficulty2)                                 0.44 1.00
## cor(difficulty2,phase2:difficulty2)                                 0.43 1.00
## cor(expected1:phase1,phase2:difficulty2)                            0.44 1.00
## cor(expected1:phase2,phase2:difficulty2)                            0.43 1.00
## cor(expected1:difficulty1,phase2:difficulty2)                       0.41 1.00
## cor(expected1:difficulty2,phase2:difficulty2)                       0.38 1.00
## cor(phase1:difficulty1,phase2:difficulty2)                          0.41 1.00
## cor(phase2:difficulty1,phase2:difficulty2)                          0.38 1.00
## cor(phase1:difficulty2,phase2:difficulty2)                          0.41 1.00
## cor(Intercept,expected1:phase1:difficulty1)                         0.44 1.00
## cor(expected1,expected1:phase1:difficulty1)                         0.38 1.00
## cor(phase1,expected1:phase1:difficulty1)                            0.40 1.00
## cor(phase2,expected1:phase1:difficulty1)                            0.46 1.00
## cor(difficulty1,expected1:phase1:difficulty1)                       0.42 1.00
## cor(difficulty2,expected1:phase1:difficulty1)                       0.41 1.00
## cor(expected1:phase1,expected1:phase1:difficulty1)                  0.41 1.00
## cor(expected1:phase2,expected1:phase1:difficulty1)                  0.43 1.00
## cor(expected1:difficulty1,expected1:phase1:difficulty1)             0.44 1.00
## cor(expected1:difficulty2,expected1:phase1:difficulty1)             0.42 1.00
## cor(phase1:difficulty1,expected1:phase1:difficulty1)                0.41 1.00
## cor(phase2:difficulty1,expected1:phase1:difficulty1)                0.39 1.00
## cor(phase1:difficulty2,expected1:phase1:difficulty1)                0.41 1.00
## cor(phase2:difficulty2,expected1:phase1:difficulty1)                0.40 1.00
## cor(Intercept,expected1:phase2:difficulty1)                         0.49 1.00
## cor(expected1,expected1:phase2:difficulty1)                         0.44 1.00
## cor(phase1,expected1:phase2:difficulty1)                            0.44 1.00
## cor(phase2,expected1:phase2:difficulty1)                            0.48 1.00
## cor(difficulty1,expected1:phase2:difficulty1)                       0.42 1.00
## cor(difficulty2,expected1:phase2:difficulty1)                       0.42 1.00
## cor(expected1:phase1,expected1:phase2:difficulty1)                  0.42 1.00
## cor(expected1:phase2,expected1:phase2:difficulty1)                  0.43 1.00
## cor(expected1:difficulty1,expected1:phase2:difficulty1)             0.42 1.00
## cor(expected1:difficulty2,expected1:phase2:difficulty1)             0.42 1.00
## cor(phase1:difficulty1,expected1:phase2:difficulty1)                0.42 1.00
## cor(phase2:difficulty1,expected1:phase2:difficulty1)                0.39 1.00
## cor(phase1:difficulty2,expected1:phase2:difficulty1)                0.42 1.00
## cor(phase2:difficulty2,expected1:phase2:difficulty1)                0.40 1.00
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1)      0.41 1.00
## cor(Intercept,expected1:phase1:difficulty2)                         0.43 1.00
## cor(expected1,expected1:phase1:difficulty2)                         0.36 1.00
## cor(phase1,expected1:phase1:difficulty2)                            0.36 1.00
## cor(phase2,expected1:phase1:difficulty2)                            0.50 1.00
## cor(difficulty1,expected1:phase1:difficulty2)                       0.41 1.00
## cor(difficulty2,expected1:phase1:difficulty2)                       0.45 1.00
## cor(expected1:phase1,expected1:phase1:difficulty2)                  0.38 1.00
## cor(expected1:phase2,expected1:phase1:difficulty2)                  0.45 1.00
## cor(expected1:difficulty1,expected1:phase1:difficulty2)             0.41 1.00
## cor(expected1:difficulty2,expected1:phase1:difficulty2)             0.45 1.00
## cor(phase1:difficulty1,expected1:phase1:difficulty2)                0.41 1.00
## cor(phase2:difficulty1,expected1:phase1:difficulty2)                0.41 1.00
## cor(phase1:difficulty2,expected1:phase1:difficulty2)                0.40 1.00
## cor(phase2:difficulty2,expected1:phase1:difficulty2)                0.41 1.00
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2)      0.41 1.00
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2)      0.43 1.00
## cor(Intercept,expected1:phase2:difficulty2)                         0.49 1.00
## cor(expected1,expected1:phase2:difficulty2)                         0.41 1.00
```

```
## cor(phase1,expected1:phase2:difficulty2)                              0.38 1.00
## cor(phase2,expected1:phase2:difficulty2)                              0.49 1.00
## cor(difficulty1,expected1:phase2:difficulty2)                         0.42 1.00
## cor(difficulty2,expected1:phase2:difficulty2)                         0.44 1.00
## cor(expected1:phase1,expected1:phase2:difficulty2)                    0.43 1.00
## cor(expected1:phase2,expected1:phase2:difficulty2)                    0.45 1.00
## cor(expected1:difficulty1,expected1:phase2:difficulty2)               0.40 1.00
## cor(expected1:difficulty2,expected1:phase2:difficulty2)               0.38 1.00
## cor(phase1:difficulty1,expected1:phase2:difficulty2)                  0.42 1.00
## cor(phase2:difficulty1,expected1:phase2:difficulty2)                  0.39 1.00
## cor(phase1:difficulty2,expected1:phase2:difficulty2)                  0.43 1.00
## cor(phase2:difficulty2,expected1:phase2:difficulty2)                  0.41 1.00
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2)        0.43 1.00
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2)        0.41 1.00
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2)        0.43 1.00
##                                                                   Bulk_ESS
## sd(Intercept)                                                         3149
## sd(expected1)                                                         4745
## sd(phase1)                                                            7678
## sd(phase2)                                                            7995
## sd(difficulty1)                                                       5760
## sd(difficulty2)                                                       6791
## sd(expected1:phase1)                                                  4821
## sd(expected1:phase2)                                                  5887
## sd(expected1:difficulty1)                                             9067
## sd(expected1:difficulty2)                                             6088
## sd(phase1:difficulty1)                                               10782
## sd(phase2:difficulty1)                                                4812
## sd(phase1:difficulty2)                                                8954
## sd(phase2:difficulty2)                                                5878
## sd(expected1:phase1:difficulty1)                                      8722
## sd(expected1:phase2:difficulty1)                                      8523
## sd(expected1:phase1:difficulty2)                                      6855
## sd(expected1:phase2:difficulty2)                                      8083
## cor(Intercept,expected1)                                             27289
## cor(Intercept,phase1)                                                 9349
## cor(expected1,phase1)                                                 2613
## cor(Intercept,phase2)                                                18825
## cor(expected1,phase2)                                                 8621
## cor(phase1,phase2)                                                   15074
## cor(Intercept,difficulty1)                                           34860
## cor(expected1,difficulty1)                                           17919
## cor(phase1,difficulty1)                                              21107
## cor(phase2,difficulty1)                                              25935
## cor(Intercept,difficulty2)                                           35309
## cor(expected1,difficulty2)                                           24094
## cor(phase1,difficulty2)                                              28374
## cor(phase2,difficulty2)                                              27923
## cor(difficulty1,difficulty2)                                         20285
## cor(Intercept,expected1:phase1)                                      35442
## cor(expected1,expected1:phase1)                                      19747
## cor(phase1,expected1:phase1)                                         30130
## cor(phase2,expected1:phase1)                                         20802
## cor(difficulty1,expected1:phase1)                                    17965
## cor(difficulty2,expected1:phase1)                                    16948
## cor(Intercept,expected1:phase2)                                      34728
## cor(expected1,expected1:phase2)                                      26274
## cor(phase1,expected1:phase2)                                         27911
## cor(phase2,expected1:phase2)                                         22263
## cor(difficulty1,expected1:phase2)                                    19179
```

```
## cor(difficulty2,expected1:phase2)                          14220
## cor(expected1:phase1,expected1:phase2)                     17477
## cor(Intercept,expected1:difficulty1)                       35774
## cor(expected1,expected1:difficulty1)                       31034
## cor(phase1,expected1:difficulty1)                          31936
## cor(phase2,expected1:difficulty1)                          27614
## cor(difficulty1,expected1:difficulty1)                     21439
## cor(difficulty2,expected1:difficulty1)                     20297
## cor(expected1:phase1,expected1:difficulty1)                19323
## cor(expected1:phase2,expected1:difficulty1)                15576
## cor(Intercept,expected1:difficulty2)                       26699
## cor(expected1,expected1:difficulty2)                       12358
## cor(phase1,expected1:difficulty2)                          18573
## cor(phase2,expected1:difficulty2)                          19170
## cor(difficulty1,expected1:difficulty2)                     12139
## cor(difficulty2,expected1:difficulty2)                     14601
## cor(expected1:phase1,expected1:difficulty2)                13726
## cor(expected1:phase2,expected1:difficulty2)                11892
## cor(expected1:difficulty1,expected1:difficulty2)           12103
## cor(Intercept,phase1:difficulty1)                          37352
## cor(expected1,phase1:difficulty1)                          32823
## cor(phase1,phase1:difficulty1)                             35470
## cor(phase2,phase1:difficulty1)                             27977
## cor(difficulty1,phase1:difficulty1)                        21646
## cor(difficulty2,phase1:difficulty1)                        19556
## cor(expected1:phase1,phase1:difficulty1)                   18786
## cor(expected1:phase2,phase1:difficulty1)                   17036
## cor(expected1:difficulty1,phase1:difficulty1)              14101
## cor(expected1:difficulty2,phase1:difficulty1)              18993
## cor(Intercept,phase2:difficulty1)                          28050
## cor(expected1,phase2:difficulty1)                          14117
## cor(phase1,phase2:difficulty1)                             25198
## cor(phase2,phase2:difficulty1)                             20726
## cor(difficulty1,phase2:difficulty1)                        13103
## cor(difficulty2,phase2:difficulty1)                        13036
## cor(expected1:phase1,phase2:difficulty1)                   13805
## cor(expected1:phase2,phase2:difficulty1)                   14182
## cor(expected1:difficulty1,phase2:difficulty1)              13134
## cor(expected1:difficulty2,phase2:difficulty1)              15978
## cor(phase1:difficulty1,phase2:difficulty1)                 12604
## cor(Intercept,phase1:difficulty2)                          34092
## cor(expected1,phase1:difficulty2)                          30039
## cor(phase1,phase1:difficulty2)                             31498
## cor(phase2,phase1:difficulty2)                             25545
## cor(difficulty1,phase1:difficulty2)                        20397
## cor(difficulty2,phase1:difficulty2)                        19287
## cor(expected1:phase1,phase1:difficulty2)                   18012
## cor(expected1:phase2,phase1:difficulty2)                   16964
## cor(expected1:difficulty1,phase1:difficulty2)              13838
## cor(expected1:difficulty2,phase1:difficulty2)              18331
## cor(phase1:difficulty1,phase1:difficulty2)                 13161
## cor(phase2:difficulty1,phase1:difficulty2)                 15761
## cor(Intercept,phase2:difficulty2)                          31526
## cor(expected1,phase2:difficulty2)                          24938
## cor(phase1,phase2:difficulty2)                             23884
## cor(phase2,phase2:difficulty2)                             25363
## cor(difficulty1,phase2:difficulty2)                        18090
## cor(difficulty2,phase2:difficulty2)                        16719
## cor(expected1:phase1,phase2:difficulty2)                   17322
## cor(expected1:phase2,phase2:difficulty2)                   15872
```

```
## cor(expected1:difficulty1,phase2:difficulty2)                          14167
## cor(expected1:difficulty2,phase2:difficulty2)                          18084
## cor(phase1:difficulty1,phase2:difficulty2)                             12081
## cor(phase2:difficulty1,phase2:difficulty2)                             14874
## cor(phase1:difficulty2,phase2:difficulty2)                             12833
## cor(Intercept,expected1:phase1:difficulty1)                            33605
## cor(expected1,expected1:phase1:difficulty1)                            27705
## cor(phase1,expected1:phase1:difficulty1)                               35093
## cor(phase2,expected1:phase1:difficulty1)                               27745
## cor(difficulty1,expected1:phase1:difficulty1)                          21512
## cor(difficulty2,expected1:phase1:difficulty1)                          20067
## cor(expected1:phase1,expected1:phase1:difficulty1)                     18415
## cor(expected1:phase2,expected1:phase1:difficulty1)                     16294
## cor(expected1:difficulty1,expected1:phase1:difficulty1)                15096
## cor(expected1:difficulty2,expected1:phase1:difficulty1)                20155
## cor(phase1:difficulty1,expected1:phase1:difficulty1)                   12484
## cor(phase2:difficulty1,expected1:phase1:difficulty1)                   15997
## cor(phase1:difficulty2,expected1:phase1:difficulty1)                   11619
## cor(phase2:difficulty2,expected1:phase1:difficulty1)                   12280
## cor(Intercept,expected1:phase2:difficulty1)                            26990
## cor(expected1,expected1:phase2:difficulty1)                            27447
## cor(phase1,expected1:phase2:difficulty1)                               30311
## cor(phase2,expected1:phase2:difficulty1)                               22611
## cor(difficulty1,expected1:phase2:difficulty1)                          20483
## cor(difficulty2,expected1:phase2:difficulty1)                          19350
## cor(expected1:phase1,expected1:phase2:difficulty1)                     17896
## cor(expected1:phase2,expected1:phase2:difficulty1)                     17484
## cor(expected1:difficulty1,expected1:phase2:difficulty1)                14938
## cor(expected1:difficulty2,expected1:phase2:difficulty1)                20080
## cor(phase1:difficulty1,expected1:phase2:difficulty1)                   12491
## cor(phase2:difficulty1,expected1:phase2:difficulty1)                   15188
## cor(phase1:difficulty2,expected1:phase2:difficulty1)                   11531
## cor(phase2:difficulty2,expected1:phase2:difficulty1)                   12837
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1)         10815
## cor(Intercept,expected1:phase1:difficulty2)                            34730
## cor(expected1,expected1:phase1:difficulty2)                            23642
## cor(phase1,expected1:phase1:difficulty2)                               28945
## cor(phase2,expected1:phase1:difficulty2)                               20249
## cor(difficulty1,expected1:phase1:difficulty2)                          19707
## cor(difficulty2,expected1:phase1:difficulty2)                          18108
## cor(expected1:phase1,expected1:phase1:difficulty2)                     15806
## cor(expected1:phase2,expected1:phase1:difficulty2)                     14364
## cor(expected1:difficulty1,expected1:phase1:difficulty2)                15408
## cor(expected1:difficulty2,expected1:phase1:difficulty2)                18049
## cor(phase1:difficulty1,expected1:phase1:difficulty2)                   13491
## cor(phase2:difficulty1,expected1:phase1:difficulty2)                   17107
## cor(phase1:difficulty2,expected1:phase1:difficulty2)                   12272
## cor(phase2:difficulty2,expected1:phase1:difficulty2)                   12874
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2)         11083
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2)         11112
## cor(Intercept,expected1:phase2:difficulty2)                            28133
## cor(expected1,expected1:phase2:difficulty2)                            28291
## cor(phase1,expected1:phase2:difficulty2)                               31934
## cor(phase2,expected1:phase2:difficulty2)                               25747
## cor(difficulty1,expected1:phase2:difficulty2)                          21361
## cor(difficulty2,expected1:phase2:difficulty2)                          18911
## cor(expected1:phase1,expected1:phase2:difficulty2)                     19111
## cor(expected1:phase2,expected1:phase2:difficulty2)                     16030
## cor(expected1:difficulty1,expected1:phase2:difficulty2)                13591
## cor(expected1:difficulty2,expected1:phase2:difficulty2)                19047
```

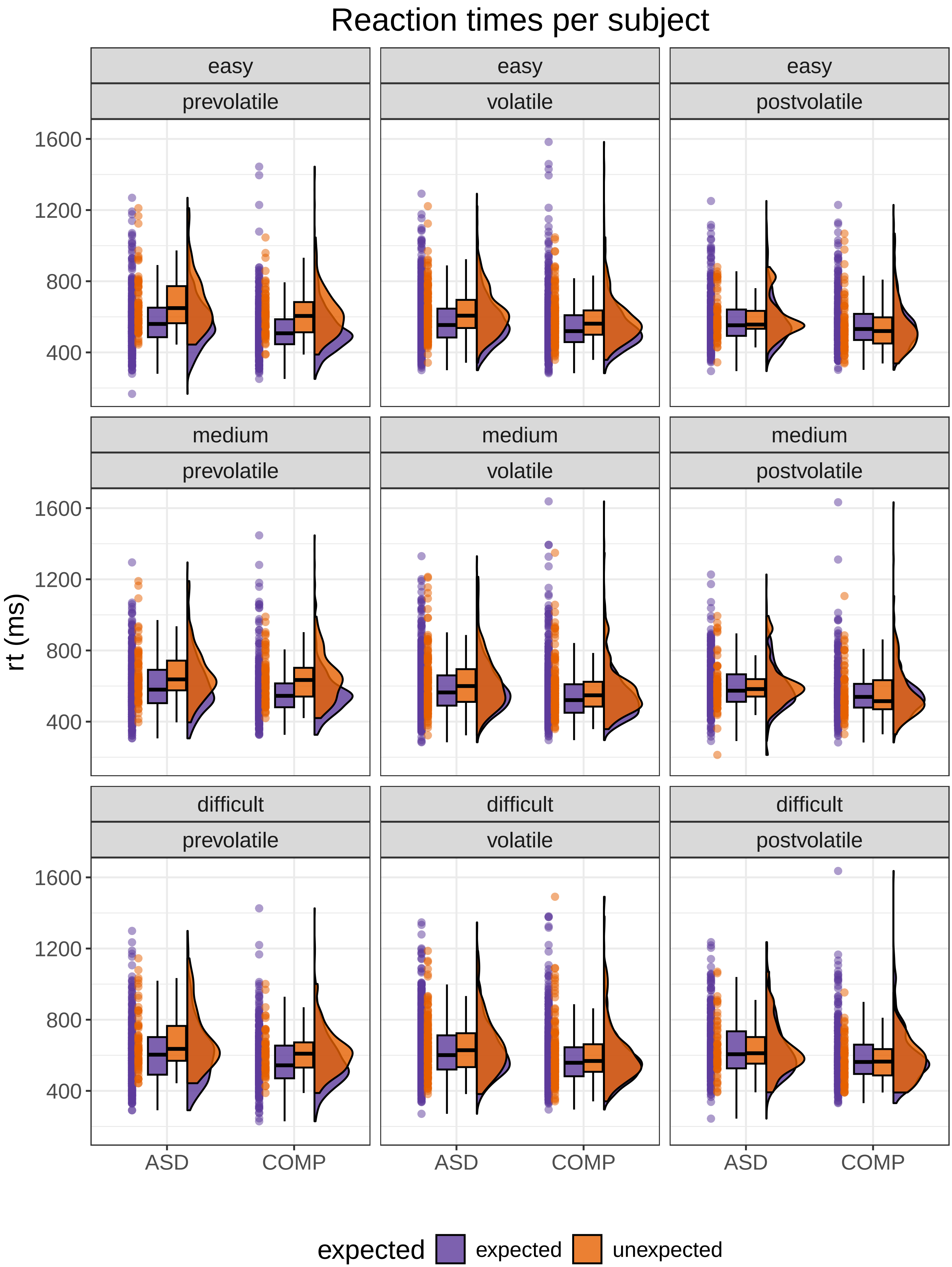
```
## cor(phase1:difficulty1,expected1:phase2:difficulty2)             12615
## cor(phase2:difficulty1,expected1:phase2:difficulty2)             14862
## cor(phase1:difficulty2,expected1:phase2:difficulty2)             12112
## cor(phase2:difficulty2,expected1:phase2:difficulty2)             13632
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2)   11191
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2)    9659
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2)   11249
##                                                                Tail_ESS
## sd(Intercept)                                                      5544
## sd(expected1)                                                      4084
## sd(phase1)                                                        11763
## sd(phase2)                                                         9176
## sd(difficulty1)                                                    7605
## sd(difficulty2)                                                    8670
## sd(expected1:phase1)                                               8183
## sd(expected1:phase2)                                               8133
## sd(expected1:difficulty1)                                          8851
## sd(expected1:difficulty2)                                          4343
## sd(phase1:difficulty1)                                             9577
## sd(phase2:difficulty1)                                             5235
## sd(phase1:difficulty2)                                             8480
## sd(phase2:difficulty2)                                             7346
## sd(expected1:phase1:difficulty1)                                   8137
## sd(expected1:phase2:difficulty1)                                   9658
## sd(expected1:phase1:difficulty2)                                   7577
## sd(expected1:phase2:difficulty2)                                   8810
## cor(Intercept,expected1)                                          14201
## cor(Intercept,phase1)                                             11917
## cor(expected1,phase1)                                              5125
## cor(Intercept,phase2)                                             14694
## cor(expected1,phase2)                                             11478
## cor(phase1,phase2)                                                14457
## cor(Intercept,difficulty1)                                        14221
## cor(expected1,difficulty1)                                        12664
## cor(phase1,difficulty1)                                           14083
## cor(phase2,difficulty1)                                           13281
## cor(Intercept,difficulty2)                                        12870
## cor(expected1,difficulty2)                                        14092
## cor(phase1,difficulty2)                                           14199
## cor(phase2,difficulty2)                                           14664
## cor(difficulty1,difficulty2)                                      13949
## cor(Intercept,expected1:phase1)                                   12829
## cor(expected1,expected1:phase1)                                   13463
## cor(phase1,expected1:phase1)                                      13750
## cor(phase2,expected1:phase1)                                      13930
## cor(difficulty1,expected1:phase1)                                 14732
## cor(difficulty2,expected1:phase1)                                 15016
## cor(Intercept,expected1:phase2)                                   13878
## cor(expected1,expected1:phase2)                                   13258
## cor(phase1,expected1:phase2)                                      13628
## cor(phase2,expected1:phase2)                                      14212
## cor(difficulty1,expected1:phase2)                                 14839
## cor(difficulty2,expected1:phase2)                                 13554
## cor(expected1:phase1,expected1:phase2)                            15153
## cor(Intercept,expected1:difficulty1)                              13338
## cor(expected1,expected1:difficulty1)                              13438
## cor(phase1,expected1:difficulty1)                                 12880
## cor(phase2,expected1:difficulty1)                                 13220
## cor(difficulty1,expected1:difficulty1)                            14011
## cor(difficulty2,expected1:difficulty1)                            14546
```

```
## cor(expected1:phase1,expected1:difficulty1)                        14257
## cor(expected1:phase2,expected1:difficulty1)                        12705
## cor(Intercept,expected1:difficulty2)                               13952
## cor(expected1,expected1:difficulty2)                               13583
## cor(phase1,expected1:difficulty2)                                  12697
## cor(phase2,expected1:difficulty2)                                  14952
## cor(difficulty1,expected1:difficulty2)                             14318
## cor(difficulty2,expected1:difficulty2)                             14570
## cor(expected1:phase1,expected1:difficulty2)                        15096
## cor(expected1:phase2,expected1:difficulty2)                        14788
## cor(expected1:difficulty1,expected1:difficulty2)                   15125
## cor(Intercept,phase1:difficulty1)                                  13678
## cor(expected1,phase1:difficulty1)                                  14233
## cor(phase1,phase1:difficulty1)                                     13191
## cor(phase2,phase1:difficulty1)                                     13129
## cor(difficulty1,phase1:difficulty1)                                13692
## cor(difficulty2,phase1:difficulty1)                                14200
## cor(expected1:phase1,phase1:difficulty1)                           14026
## cor(expected1:phase2,phase1:difficulty1)                           14219
## cor(expected1:difficulty1,phase1:difficulty1)                      14254
## cor(expected1:difficulty2,phase1:difficulty1)                      15218
## cor(Intercept,phase2:difficulty1)                                  12762
## cor(expected1,phase2:difficulty1)                                  13558
## cor(phase1,phase2:difficulty1)                                     14444
## cor(phase2,phase2:difficulty1)                                     13904
## cor(difficulty1,phase2:difficulty1)                                13969
## cor(difficulty2,phase2:difficulty1)                                14586
## cor(expected1:phase1,phase2:difficulty1)                           14625
## cor(expected1:phase2,phase2:difficulty1)                           15007
## cor(expected1:difficulty1,phase2:difficulty1)                      13761
## cor(expected1:difficulty2,phase2:difficulty1)                      16048
## cor(phase1:difficulty1,phase2:difficulty1)                         14648
## cor(Intercept,phase1:difficulty2)                                  13578
## cor(expected1,phase1:difficulty2)                                  12528
## cor(phase1,phase1:difficulty2)                                     13553
## cor(phase2,phase1:difficulty2)                                     12976
## cor(difficulty1,phase1:difficulty2)                                13062
## cor(difficulty2,phase1:difficulty2)                                13990
## cor(expected1:phase1,phase1:difficulty2)                           14922
## cor(expected1:phase2,phase1:difficulty2)                           15245
## cor(expected1:difficulty1,phase1:difficulty2)                      13665
## cor(expected1:difficulty2,phase1:difficulty2)                      14660
## cor(phase1:difficulty1,phase1:difficulty2)                         14547
## cor(phase2:difficulty1,phase1:difficulty2)                         15387
## cor(Intercept,phase2:difficulty2)                                  13859
## cor(expected1,phase2:difficulty2)                                  14301
## cor(phase1,phase2:difficulty2)                                     12800
## cor(phase2,phase2:difficulty2)                                     14350
## cor(difficulty1,phase2:difficulty2)                                14777
## cor(difficulty2,phase2:difficulty2)                                14355
## cor(expected1:phase1,phase2:difficulty2)                           14374
## cor(expected1:phase2,phase2:difficulty2)                           15093
## cor(expected1:difficulty1,phase2:difficulty2)                      15077
## cor(expected1:difficulty2,phase2:difficulty2)                      15242
## cor(phase1:difficulty1,phase2:difficulty2)                         14853
## cor(phase2:difficulty1,phase2:difficulty2)                         15856
## cor(phase1:difficulty2,phase2:difficulty2)                         14907
## cor(Intercept,expected1:phase1:difficulty1)                        13388
## cor(expected1,expected1:phase1:difficulty1)                        14077
## cor(phase1,expected1:phase1:difficulty1)                           13484
```

```
## cor(phase2,expected1:phase1:difficulty1)                          13443
## cor(difficulty1,expected1:phase1:difficulty1)                     14630
## cor(difficulty2,expected1:phase1:difficulty1)                     13979
## cor(expected1:phase1,expected1:phase1:difficulty1)                15474
## cor(expected1:phase2,expected1:phase1:difficulty1)                14892
## cor(expected1:difficulty1,expected1:phase1:difficulty1)           15168
## cor(expected1:difficulty2,expected1:phase1:difficulty1)           14844
## cor(phase1:difficulty1,expected1:phase1:difficulty1)              14111
## cor(phase2:difficulty1,expected1:phase1:difficulty1)              15451
## cor(phase1:difficulty2,expected1:phase1:difficulty1)              13985
## cor(phase2:difficulty2,expected1:phase1:difficulty1)              14042
## cor(Intercept,expected1:phase2:difficulty1)                       13634
## cor(expected1,expected1:phase2:difficulty1)                       13657
## cor(phase1,expected1:phase2:difficulty1)                          13391
## cor(phase2,expected1:phase2:difficulty1)                          13987
## cor(difficulty1,expected1:phase2:difficulty1)                     13551
## cor(difficulty2,expected1:phase2:difficulty1)                     14253
## cor(expected1:phase1,expected1:phase2:difficulty1)                14080
## cor(expected1:phase2,expected1:phase2:difficulty1)                14275
## cor(expected1:difficulty1,expected1:phase2:difficulty1)           14949
## cor(expected1:difficulty2,expected1:phase2:difficulty1)           15395
## cor(phase1:difficulty1,expected1:phase2:difficulty1)              14651
## cor(phase2:difficulty1,expected1:phase2:difficulty1)              14521
## cor(phase1:difficulty2,expected1:phase2:difficulty1)              13935
## cor(phase2:difficulty2,expected1:phase2:difficulty1)              14099
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty1)    14314
## cor(Intercept,expected1:phase1:difficulty2)                       12487
## cor(expected1,expected1:phase1:difficulty2)                       14044
## cor(phase1,expected1:phase1:difficulty2)                          13872
## cor(phase2,expected1:phase1:difficulty2)                          13094
## cor(difficulty1,expected1:phase1:difficulty2)                     13371
## cor(difficulty2,expected1:phase1:difficulty2)                     14929
## cor(expected1:phase1,expected1:phase1:difficulty2)                15211
## cor(expected1:phase2,expected1:phase1:difficulty2)                14118
## cor(expected1:difficulty1,expected1:phase1:difficulty2)           15059
## cor(expected1:difficulty2,expected1:phase1:difficulty2)           15279
## cor(phase1:difficulty1,expected1:phase1:difficulty2)              15251
## cor(phase2:difficulty1,expected1:phase1:difficulty2)              15467
## cor(phase1:difficulty2,expected1:phase1:difficulty2)              14909
## cor(phase2:difficulty2,expected1:phase1:difficulty2)              14660
## cor(expected1:phase1:difficulty1,expected1:phase1:difficulty2)    14105
## cor(expected1:phase2:difficulty1,expected1:phase1:difficulty2)    14394
## cor(Intercept,expected1:phase2:difficulty2)                       13149
## cor(expected1,expected1:phase2:difficulty2)                       13706
## cor(phase1,expected1:phase2:difficulty2)                          13551
## cor(phase2,expected1:phase2:difficulty2)                          13922
## cor(difficulty1,expected1:phase2:difficulty2)                     14221
## cor(difficulty2,expected1:phase2:difficulty2)                     14651
## cor(expected1:phase1,expected1:phase2:difficulty2)                14390
## cor(expected1:phase2,expected1:phase2:difficulty2)                14380
## cor(expected1:difficulty1,expected1:phase2:difficulty2)           14311
## cor(expected1:difficulty2,expected1:phase2:difficulty2)           15488
## cor(phase1:difficulty1,expected1:phase2:difficulty2)              14330
## cor(phase2:difficulty1,expected1:phase2:difficulty2)              13690
## cor(phase1:difficulty2,expected1:phase2:difficulty2)              13794
## cor(phase2:difficulty2,expected1:phase2:difficulty2)              15011
## cor(expected1:phase1:difficulty1,expected1:phase2:difficulty2)    15334
## cor(expected1:phase2:difficulty1,expected1:phase2:difficulty2)    13620
## cor(expected1:phase1:difficulty2,expected1:phase2:difficulty2)    13966
##
```

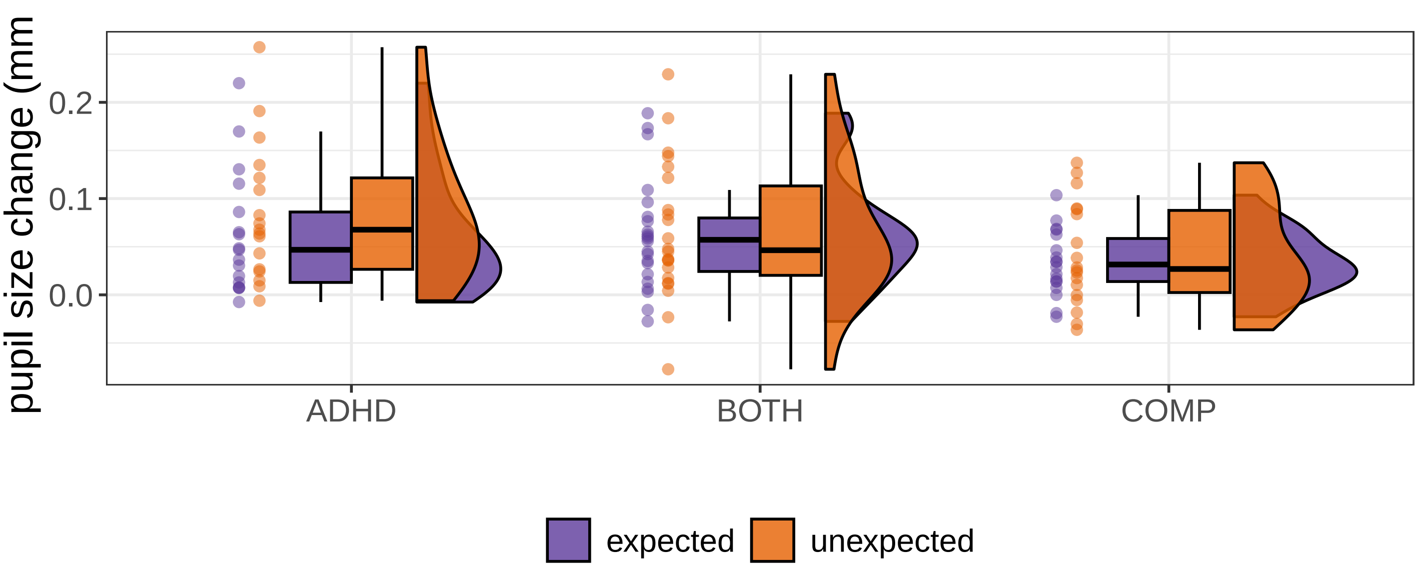
```
## ~trl (Number of levels: 288)
##                           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## sd(Intercept)                 0.06      0.00     0.05     0.07 1.00     7395
## sd(diagnosis1)                0.01      0.00     0.00     0.02 1.00     3741
## cor(Intercept,diagnosis1)     0.03      0.35    -0.69     0.71 1.00    21237
##                           Tail_ESS
## sd(Intercept)                12182
## sd(diagnosis1)                7318
## cor(Intercept,diagnosis1)    10591
##
## Regression Coefficients:
##                                          Estimate Est.Error l-95% CI u-95% CI
## Intercept                                    6.17      0.03     6.12     6.23
## diagnosis1                                   0.03      0.02    -0.01     0.07
## expected1                                   -0.04      0.01    -0.05    -0.02
## phase1                                       0.03      0.01     0.00     0.05
## phase2                                      -0.01      0.01    -0.02     0.01
## difficulty1                                 -0.03      0.01    -0.04    -0.01
## difficulty2                                 -0.01      0.01    -0.02     0.01
## diagnosis1:expected1                        -0.00      0.00    -0.01     0.01
## diagnosis1:phase1                            0.00      0.01    -0.02     0.02
## diagnosis1:phase2                           -0.00      0.01    -0.01     0.01
## expected1:phase1                            -0.04      0.01    -0.05    -0.02
## expected1:phase2                             0.00      0.01    -0.01     0.02
## diagnosis1:difficulty1                       0.00      0.00    -0.01     0.01
## diagnosis1:difficulty2                      -0.00      0.00    -0.01     0.01
## expected1:difficulty1                       -0.01      0.01    -0.03     0.00
## expected1:difficulty2                        0.00      0.01    -0.01     0.02
## phase1:difficulty1                          -0.00      0.01    -0.02     0.02
## phase2:difficulty1                           0.01      0.01    -0.01     0.03
## phase1:difficulty2                           0.02      0.01    -0.00     0.04
## phase2:difficulty2                          -0.01      0.01    -0.03     0.01
## diagnosis1:expected1:phase1                  0.01      0.01    -0.00     0.02
## diagnosis1:expected1:phase2                  0.00      0.00    -0.01     0.01
## diagnosis1:expected1:difficulty1            -0.00      0.00    -0.01     0.01
## diagnosis1:expected1:difficulty2             0.00      0.00    -0.01     0.01
## diagnosis1:phase1:difficulty1                0.01      0.01    -0.01     0.02
## diagnosis1:phase2:difficulty1               -0.00      0.01    -0.01     0.01
## diagnosis1:phase1:difficulty2               -0.01      0.01    -0.02     0.01
## diagnosis1:phase2:difficulty2                0.01      0.01    -0.01     0.02
## expected1:phase1:difficulty1                -0.01      0.01    -0.03     0.01
## expected1:phase2:difficulty1                -0.00      0.01    -0.02     0.02
## expected1:phase1:difficulty2                 0.01      0.01    -0.02     0.03
## expected1:phase2:difficulty2                -0.00      0.01    -0.02     0.02
## diagnosis1:expected1:phase1:difficulty1      0.00      0.01    -0.01     0.02
## diagnosis1:expected1:phase2:difficulty1      0.00      0.01    -0.01     0.01
## diagnosis1:expected1:phase1:difficulty2     -0.00      0.01    -0.01     0.01
## diagnosis1:expected1:phase2:difficulty2     -0.00      0.01    -0.02     0.01
##                                          Rhat Bulk_ESS Tail_ESS
## Intercept                                1.00     1845     4403
## diagnosis1                               1.00     1884     3305
## expected1                                1.00    10083    12702
## phase1                                   1.00     6761     9589
## phase2                                   1.00     8441    11148
## difficulty1                              1.00     8289    11626
## difficulty2                              1.00     8504    11731
## diagnosis1:expected1                     1.00    19211    14759
## diagnosis1:phase1                        1.00     6434    10799
## diagnosis1:phase2                        1.00    15067    14633
## expected1:phase1                         1.00     9181    11343
```

```
## expected1:phase2                          1.00     8635    12070
## diagnosis1:difficulty1                    1.00    16956    14624
## diagnosis1:difficulty2                    1.00    18692    15227
## expected1:difficulty1                     1.00     8002    11088
## expected1:difficulty2                     1.00     7965    10283
## phase1:difficulty1                        1.00     8060    11648
## phase2:difficulty1                        1.00     7989    11209
## phase1:difficulty2                        1.00     8367    11425
## phase2:difficulty2                        1.00     7970    11218
## diagnosis1:expected1:phase1               1.00    25920    15621
## diagnosis1:expected1:phase2               1.00    23007    15510
## diagnosis1:expected1:difficulty1          1.00    17794    14751
## diagnosis1:expected1:difficulty2          1.00    16122    14948
## diagnosis1:phase1:difficulty1             1.00    15240    14451
## diagnosis1:phase2:difficulty1             1.00    13895    12915
## diagnosis1:phase1:difficulty2             1.00    15980    14790
## diagnosis1:phase2:difficulty2             1.00    15110    14708
## expected1:phase1:difficulty1              1.00     7790    11261
## expected1:phase2:difficulty1              1.00     7602    11295
## expected1:phase1:difficulty2              1.00     8608    12268
## expected1:phase2:difficulty2              1.00     8034    10958
## diagnosis1:expected1:phase1:difficulty1 1.00    15527    13584
## diagnosis1:expected1:phase2:difficulty1 1.00    14325    14873
## diagnosis1:expected1:phase1:difficulty2 1.00    16115    14760
## diagnosis1:expected1:phase2:difficulty2 1.00    15825    14645
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.23      0.00     0.22     0.24 1.00    13158    13192
## ndt     100.15      7.50    84.81   114.20 1.00    12727    13392
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```


Reaction times per subject
easy
prevolatile
easy
volatile
easy
postvolatile
medium
prevolatile
medium
volatile
medium
postvolatile
difficult
prevolatile
difficult
volatile
difficult
postvolatile
rt (ms)
1600
1200
800
400
ASD
COMP
expected
expected
unexpected

S4.3 Pupil sizes

```
##  Family: gaussian
##   Links: mu = identity
## Formula: rel_pupil ~ diagnosis * expected + rts + (1 | subID)
##    Data: df (Number of observations: 14661)
##   Draws: 4 chains, each with iter = 6000; warmup = 1500; thin = 1;
##          total post-warmup draws = 18000
##
## Multilevel Hyperparameters:
## ~subID (Number of levels: 57)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     0.06      0.01     0.05     0.07 1.00     1831     3420
##
## Regression Coefficients:
##                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                0.06      0.01     0.04     0.07 1.01     1301
## diagnosis1               0.01      0.01    -0.01     0.03 1.00     1276
## diagnosis2               0.00      0.01    -0.02     0.03 1.00     1087
## expected1               -0.00      0.00    -0.01     0.00 1.00    38787
## rts                      0.00      0.00     0.00     0.00 1.00    20007
## diagnosis1:expected1    -0.00      0.00    -0.01     0.00 1.00    21529
## diagnosis2:expected1     0.00      0.00    -0.00     0.01 1.00    21870
##                      Tail_ESS
## Intercept                2673
## diagnosis1               2062
## diagnosis2               1993
## expected1               12036
## rts                     15436
## diagnosis1:expected1    14814
## diagnosis2:expected1    13960
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.18      0.00     0.17     0.18 1.00    20066    11972
##
## Draws were sampled using sample(hmc). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```



S4.4 Accuracies

```
## ---
```

```
##  Model:
## Type: BFlinearModel, JZS
## Intercept only
## Data types:
## diagnosis :  fixed
```

	bf
diagnosis + expected + difficulty	46.41
diagnosis + difficulty	46.25
diagnosis + expected + difficulty + diagnosis:difficulty	45.02
diagnosis + difficulty + diagnosis:difficulty	44.85
diagnosis + phase + expected + phase:expected + difficulty	44.77
expected + difficulty	44.42
difficulty	44.28
diagnosis + expected + difficulty + expected:difficulty	43.35
diagnosis + phase + expected + difficulty	43.23
diagnosis + phase + expected + phase:expected + difficulty + diagnosis:difficulty	43.19
diagnosis + expected + diagnosis:expected + difficulty	43.15
diagnosis + phase + difficulty	43.00
phase + expected + phase:expected + difficulty	42.44
diagnosis + expected + difficulty + diagnosis:difficulty + expected:difficulty	42.04
diagnosis + phase + expected + difficulty + diagnosis:difficulty	41.95
diagnosis + expected + diagnosis:expected + difficulty + diagnosis:difficulty	41.72
diagnosis + phase + difficulty + diagnosis:difficulty	41.71
diagnosis + phase + expected + phase:expected + difficulty + expected:difficulty	41.53
expected + difficulty + expected:difficulty	41.35
phase + expected + difficulty	41.20
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty	41.20
phase + difficulty	41.03
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty	40.62
diagnosis + phase + expected + phase:expected + difficulty + diagnosis:difficulty + expected:difficulty	40.26
diagnosis + phase + expected + difficulty + expected:difficulty	40.25
diagnosis + expected + diagnosis:expected + difficulty + expected:difficulty	39.94
diagnosis + phase + expected + diagnosis:expected + difficulty	39.85
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty	39.84
diagnosis + phase + expected + phase:expected + difficulty + phase:difficulty	39.46
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty	39.35
phase + expected + phase:expected + difficulty + expected:difficulty	39.33
diagnosis + phase + diagnosis:phase + expected + difficulty	39.25
diagnosis + phase + diagnosis:phase + difficulty	39.13
diagnosis + phase + expected + difficulty + diagnosis:difficulty + expected:difficulty	38.86
diagnosis + expected + diagnosis:expected + difficulty + diagnosis:difficulty + expected:difficulty	38.69
diagnosis + phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty	38.62
diagnosis + phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty	38.43
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + expected:difficulty	38.13
phase + expected + difficulty + expected:difficulty	38.13
diagnosis + phase + expected + difficulty + phase:difficulty	38.09
diagnosis + phase + diagnosis:phase + expected + difficulty + diagnosis:difficulty	38.05
diagnosis + phase + difficulty + phase:difficulty	38.04
diagnosis + phase + diagnosis:phase + difficulty + diagnosis:difficulty	37.77
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + expected:difficulty	37.59
phase + expected + phase:expected + difficulty + phase:difficulty	37.37
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty	37.29
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + expected:difficulty	37.02
diagnosis + phase + expected + phase:expected + difficulty + phase:difficulty + expected:difficulty	36.80
diagnosis + phase + expected + difficulty + diagnosis:difficulty + phase:difficulty	36.78
diagnosis + phase + expected + diagnosis:expected + difficulty + expected:difficulty	36.78
diagnosis + phase + difficulty + diagnosis:difficulty + phase:difficulty	36.54

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diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty + expected:difficulty	36.33
diagnosis + phase + diagnosis:phase + expected + difficulty + expected:difficulty	36.27
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + phase:difficulty	36.10
phase + expected + difficulty + phase:difficulty	36.08
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty	36.01
phase + difficulty + phase:difficulty	35.92
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty	35.92
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + phase:difficulty	35.64
diagnosis + phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + expected:difficulty	35.45
diagnosis + phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	35.20
diagnosis + phase + expected + difficulty + phase:difficulty + expected:difficulty	35.06
diagnosis + phase + expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	34.99
diagnosis + phase + diagnosis:phase + expected + difficulty + diagnosis:difficulty + expected:difficulty	34.87
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty	34.76
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty	34.74
diagnosis + phase + expected + diagnosis:expected + difficulty + phase:difficulty	34.73
phase + expected + phase:expected + difficulty + phase:difficulty + expected:difficulty	34.55
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty	34.29
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + expected:difficulty	34.25
diagnosis + phase + diagnosis:phase + expected + difficulty + phase:difficulty	34.20
diagnosis + expected + diagnosis:expected + difficulty + diagnosis:difficulty + expected:difficulty + diagnosis:expected:difficulty	34.16
diagnosis + phase + diagnosis:phase + difficulty + phase:difficulty	34.01
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty	33.86
diagnosis + phase + expected + phase:expected + difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	33.49
diagnosis + phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty	33.34
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + expected:difficulty	33.06
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + phase:difficulty + expected:difficulty	33.05
phase + expected + difficulty + phase:difficulty + expected:difficulty	33.00
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + expected:difficulty	32.91
diagnosis + phase + diagnosis:phase + expected + difficulty + diagnosis:difficulty + phase:difficulty	32.83
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty	32.74
diagnosis + phase + diagnosis:phase + difficulty + diagnosis:difficulty + phase:difficulty	32.65
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + phase:difficulty + expected:difficulty	32.59
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + expected:difficulty + diagnosis:expected:difficulty	32.34
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + phase:difficulty	32.29
diagnosis + phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	32.08
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	31.76
diagnosis + phase + expected + diagnosis:expected + difficulty + phase:difficulty + expected:difficulty	31.69
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + expected:difficulty	31.52
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	31.34
phase + expected + phase:expected + difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	31.31
diagnosis + phase + diagnosis:phase + expected + difficulty + phase:difficulty + expected:difficulty	31.15

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diagnosis + phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + expected:difficulty + diagnosis:expected:difficulty	30.95
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty	30.90
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + expected:difficulty	30.81
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + phase:difficulty	30.76
diagnosis + phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	30.45
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	29.98
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty	29.88
diagnosis + phase + diagnosis:phase + expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	29.86
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	29.56
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty	29.54
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + expected:difficulty	29.50
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + phase:difficulty + expected:difficulty	29.13
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + phase:difficulty	28.75
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	28.68
diagnosis + phase + diagnosis:phase + expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty	28.48
diagnosis + phase + diagnosis:phase + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty	28.19
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	28.17
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	28.15
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + expected:difficulty + diagnosis:expected:difficulty	27.81
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + phase:difficulty + expected:difficulty	27.80
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty	27.54
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	27.31
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + expected:difficulty + diagnosis:expected:difficulty	27.03
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty	26.76
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty	26.57
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	26.42
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	26.17
diagnosis + phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	25.83
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + phase:difficulty + expected:difficulty	25.81
diagnosis + phase + diagnosis:phase + expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty	25.42

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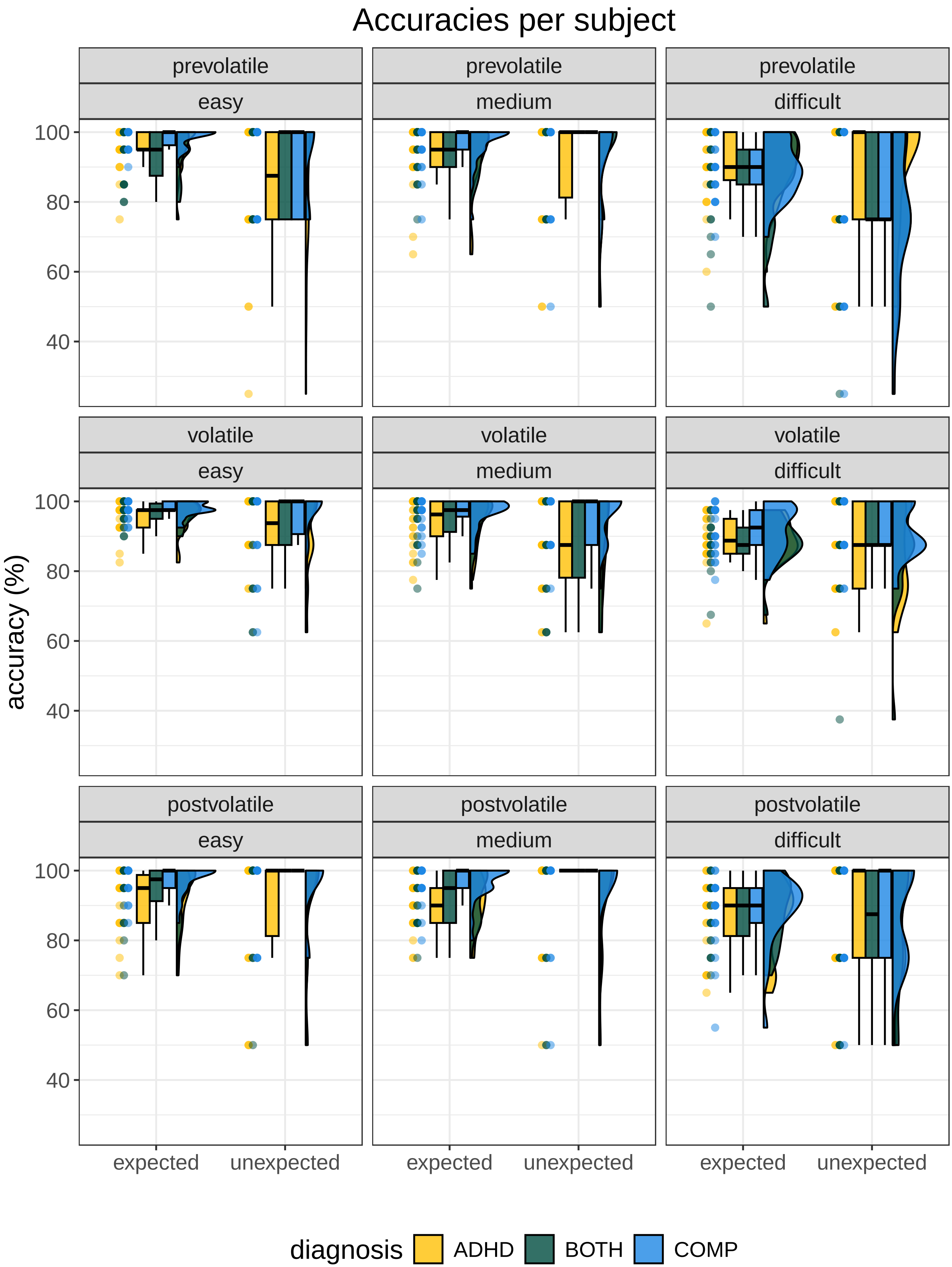
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty	25.11
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + expected:difficulty + diagnosis:expected:difficulty	25.08
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	24.94
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty	24.67
diagnosis + phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty + phase:expected:difficulty	24.26
diagnosis + phase + diagnosis:phase + expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + phase:expected:difficulty	23.78
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty	23.49
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	23.46
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty	23.04
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	22.73
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty	22.03
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	22.02
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + phase:expected:difficulty	21.45
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + phase:expected:difficulty	20.57
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty + phase:expected:difficulty	20.36
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty	20.07
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	19.99
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	19.07
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	17.60
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + phase:expected:difficulty	17.04
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + expected:difficulty + diagnosis:expected:difficulty + phase:expected:difficulty	16.91
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty + phase:expected:difficulty	16.05
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty	15.66
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty + phase:expected:difficulty	12.66
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected + difficulty + diagnosis:difficulty + phase:difficulty + diagnosis:phase:difficulty + expected:difficulty + diagnosis:expected:difficulty + phase:expected:difficulty + diagnosis:phase:expected:difficulty	8.72
diagnosis	1.45
diagnosis + expected	1.41

	bf
expected	-0.08
diagnosis + phase + expected + phase:expected	-1.05
diagnosis + phase	-1.92
diagnosis + phase + expected	-1.99
diagnosis + expected + diagnosis:expected	-2.00
phase + expected + phase:expected	-2.60
phase	-3.36
phase + expected	-3.46
diagnosis + phase + expected + diagnosis:expected + phase:expected	-4.40
diagnosis + phase + diagnosis:phase + expected + phase:expected	-5.17
diagnosis + phase + expected + diagnosis:expected	-5.19
diagnosis + phase + diagnosis:phase	-6.00
diagnosis + phase + diagnosis:phase + expected	-6.01
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected	-8.52
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected	-9.36
diagnosis + phase + diagnosis:phase + expected + diagnosis:expected + phase:expected + diagnosis:phase:expected	-12.14


Accuracies per subject
prevolatile
easy
prevolatile
medium
prevolatile
difficult
volatile
easy
volatile
medium
volatile
difficult
postvolatile
easy
postvolatile
medium
postvolatile
difficult
accuracy (%)
100
80
60
40
expected
unexpected
diagnosis
ADHD
BOTH
COMP